

Hybrid Microsoft Network

Build & Troubleshooting Playbook
On-Premises Active Directory + Microsoft Entra ID + Intune

Field reference for SMB and MSP environments
Derived from the MockBiz hybrid lab build

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Part 0 — How to Use This Playbook

This document has two jobs.

Job one: build a network from nothing. Parts 3 and 4 walk the entire greenfield build in the order you should actually do it — not the order you learn it in. Follow it top to bottom and you will not have to go back and redo anything.

Job two: walk into somebody else's network and figure out what's wrong. Part 5 is the assessment and triage half. That is the more common situation in MSP work: you inherit an environment built by someone who left three years ago, with no documentation, and you have to find out what's actually running before you can fix anything.

The organizing principle

Everything that shapes a user's experience gets built **before** the users exist and before any client machine is joined.

OU structure, security groups, file shares and their permissions, Group Policy objects, the UPN suffix, and the MDM enrollment policy are all foundation. If you build them after you have users and machines, every change requires you to log into client machines, log off, log back on, clear caches, and retest. If you build them first, the first user who logs in gets a finished environment and you never touch that machine again.

This is the single largest efficiency difference between a first build and a fifth build.

How each phase is written

Element	What it gives you
Do this	The literal actions, in order. Commands are given in full.
Why (green)	The reasoning. Read this — it is what you need when a client or a senior engineer asks why you did something, and it is what makes the step transferable to a different environment.
Warning (amber)	A trap that is easy to walk into and annoying to walk back out of.
Danger (red)	Something that causes real damage — data loss, lockout, or an outage. Read before acting.
Verify (blue)	How you prove the step worked before moving on. Do not skip these. Most of the painful troubleshooting in any build comes from three phases stacked on top of an unverified one.

Conventions used in the examples

Placeholder	Meaning	Example value
ad.contoso.com	Internal AD DNS domain	A subdomain of a domain the client owns
CONTOSO	NetBIOS domain name	Short, uppercase, no spaces
SITE-DC01 / SITE-DC02	Domain controllers	RC-DC01
10.10.10.0/24	Server / client subnet	Whatever the client uses
OU=Contoso,DC=ad,DC=contoso,DC=com	Top-level organization OU	Everything you build lives under here

Part 1 — Project Record

Two builds of the same environment. Round 1 was exploratory and out of order; Round 2 followed this playbook as a checklist. The comparison is the point.

Round 2 architecture (current)

Component	Detail
Public domain	homebiz.beer — registered, verified in Entra, set primary
AD DNS domain	ad.homebiz.beer — subdomain of the routable domain, no split-brain DNS
NetBIOS	HOMEBIZ
Network	Lab VLAN, 192.168.40.0/24 , gateway .1 , DHCP on the firewall with options 6 and 15
HL-DC01	HP ProDesk 400 G4 DM, bare metal, .10 . AD DS, DNS, DHCP-adjacent, file server, Entra Connect. 32 GB RAM. 512 GB NVMe (OS) + 1 TB SATA SSD as S: for shares.
HL-DC02	Hyper-V VM, .20 . AD DS, DNS, Global Catalog, replicating from DC01.
Clients	WS-ACCT01 and WS-SALES01 — Windows 11 Enterprise, domain joined, hybrid Entra joined, Intune enrolled
Cloud tenant	Microsoft 365 Business Premium with Copilot, 5 seats
Admin model	Cloud-only break-glass on .onmicrosoft.com (Owner in Azure) plus a separate daily admin on the custom domain (Contributor)
Identity bridge	Entra Connect Sync, Password Hash Sync, Seamless SSO, OU-filtered to Employees / Groups / Workstations
Backup	System State to external disk, scheduled nightly

Delivered in Round 2

- Forest, domain, DNS with forwarders, reverse lookup zone, and scavenging
- External NTP anchored on the PDC emulator
- Scheduled System State backup and AD Recycle Bin
- DHCP options delivering the DC as DNS and the AD suffix
- Single-organization OU tree with redircmp and redirusr
- Full AGDLP group model — role groups nested into resource groups
- Departmental shares with Access-Based Enumeration and group-based NTFS
- Five purpose-named GPOs, each linked to the OU holding the objects it targets
- Drive maps with item-level targeting, Folder Redirection, home folders
- Scripted bulk user creation with unique generated passwords
- Hybrid identity: routable UPN suffix, Entra Connect, SCP, hybrid join, PRT
- Intune enrollment by GPO with Corporate device ownership
- Microsoft 365 Apps deployed to a dynamic device group
- Second domain controller promoted and replicating

Round 1 vs Round 2

Dimension	Round 1	Round 2
Domain naming	.local — required a UPN rewrite later	Routable subdomain, correct from the start
Order of work	Discovery order; constant rework	Build order; no rework
OU structure	Collided with built-in containers	Single org OU, container redirection applied
GPO scoping	Linked to a groups OU — never applied	Linked where the objects live — applied first time
Permissions	Two lockouts	No lockouts; ABE and ILT verified negatively
Drive maps	Replace action; froze both clients	Update action; no incident
Device ownership	Personal — crippled one machine permanently	Corporate via GPO
Admin accounts	Single global admin used for everything	Break-glass separated from daily admin
Backup / NTP / Recycle Bin	Never done	Done in Phase 2–3
Total elapsed	Several days	Under six hours

Carried forward

See Appendix E for the current open-items list.

Part 2 — Root-Cause Findings and Corrections

This is the most valuable part of the document. Each finding below is a real failure from the build, traced to its actual cause. Several were fixed at the time with a workaround rather than a fix — those are corrected here.

Read this section before rebuilding. Most of Part 3 exists because of what is in this section.

Finding 1 — The `.local` domain name created a cascading tax

What happened: The domain was created as `mockbiz.local`. Later, when connecting to Entra ID, the wizard reported that the UPN suffix could not be verified. The workaround was to add an alternative routable UPN suffix in Active Directory Domains and Trusts and bulk-rewrite every user's UPN.

Actual cause: `.local` is reserved by RFC 6762 for multicast DNS. It is not a real top-level domain, it cannot be registered, and it therefore can never be verified in Entra ID. Microsoft has advised against it since the Server 2008 era.

WHY THIS MATTERS BEYOND THE LAB

The UPN rewrite was not an unavoidable step in hybrid identity — it was the interest payment on a bad naming decision made at hour one. Choose `ad.contoso.com` or `corp.contoso.com`, a subdomain of a routable domain the client actually owns, and that entire step disappears.

Additional costs of `.local`: public certificate authorities will not issue certificates for it, it collides with Bonjour/Avahi service discovery on macOS, iOS, printers, and Linux, and it forces split-brain DNS workarounds later.

YOU WILL INHERIT THIS

A large share of SMB domains in the field are `.local`. Renaming an AD domain is possible (`rendom`) but disruptive enough that the standard answer is to live with it and add a routable UPN suffix — exactly the workaround performed here. Know the workaround, know why it's needed, and never create a new one.

Finding 2 — Top-level OUs named "Users" and "Computers" cannot be created

What happened: The recommended OU structure could not be built. This was discovered late — after users had been created and a GPO had already been linked — and it was the root of the drive-mapping failure in Finding 3.

Actual cause: `CN=Users` and `CN=Computers` already exist at the domain root as **containers**, not organizational units. The name collision blocks creation, and the containers themselves have two hard limitations: you cannot create OUs inside them, and **you cannot link a Group Policy Object to them**.

Correction: Use one top-level OU named for the organization, with everything beneath it. Never reuse the built-in container names.

```
ad.contoso.com
├── Domain Controllers      (built-in – leave alone)
├── CN=Users                (built-in container – no GPOs reach here)
├── CN=Computers           (built-in container – no GPOs reach here)
└── OU=Contoso             ← everything you build
    ├── OU=Employees
    ├── OU=Workstations
    ├── OU=Servers
    ├── OU=Groups
    ├── OU=ServiceAccounts
    └── OU=Disabled
```

THE FIX NOBODY APPLIES

By default, every machine that joins the domain lands in `CN=Computers`, where no GPO can reach it. Run `redircmp` and `redirusr` once, immediately after building the OU tree, and new objects land somewhere policy actually applies. This eliminates an entire recurring class of "the new PC didn't get its settings" tickets.

Finding 3 — Group Policy linked to an OU containing groups

What happened: The drive-mapping GPO was linked to `Groups → Security`. It never appeared in `gpresult /r` and no drives mapped.

Actual cause: Group Policy applies based on where the **user or computer object** sits in the OU tree. It does not apply to groups and it does not care where groups live.

The rule, stated once so it sticks: OUs scope policy. Groups filter policy. Linking a GPO to an OU full of groups accomplishes nothing.

Finding 4 — Share and NTFS permission cleanup caused two lockouts

What happened: Users could see a share but not open files. A permissions cleanup was performed. Access then broke completely for everyone, including administrators.

Actual cause: Two compounding errors. First, security was being enforced at both the share layer and the NTFS layer, so every symptom had two possible causes. Second, the cleanup stripped the ACL and rebuilt it, removing `SYSTEM` and administrative access in the process.

Correction — the standard model:

Layer	Setting	Reasoning
Share permissions	<code>Authenticated Users</code> = Full Control. Remove <code>Everyone</code> .	Opened deliberately so it is never the cause of a problem
NTFS permissions	All real access control, assigned to groups	One place to look, one place to fix

TWO RULES THAT PROTECT YOU

Never remove `SYSTEM` or the administrative group from an ACL. The operating system needs `SYSTEM`. Removing it can lock the server out of its own data, and recovery means taking ownership and rebuilding from scratch.

"Replace all child object permission entries" is destructive and irreversible. It erases every explicit permission beneath the folder. On a new empty tree it is harmless. On a client's production file server with a decade of accumulated per-folder permissions, it destroys the entire permission model with no undo.

Finding 5 — Folder Redirection Event ID 502, "Access is denied"

What happened: Folder Redirection was configured correctly, the GPO applied, permissions were verified repeatedly, and redirection still failed. It was eventually resolved by unchecking "Grant the user exclusive rights."

Actual cause: Home folders and redirected folders were pointed at the same root path. Active Directory's home-folder feature creates the user's folder when you click Apply — and the **owner** of that folder is the administrator account that created it, not the user. Folder Redirection with "Grant exclusive rights" enabled then checks ownership of the existing folder, finds it does not belong to the user, and refuses with Access Denied.

Correction: Use separate roots. `\\SERVER\Home` for home folders, `\\SERVER\Redirected` for Folder Redirection. The conflict then cannot occur and you keep the tighter per-user ACL that exclusive rights provides.

WORKAROUND VS FIX

Unchecking "Grant exclusive rights" made redirection work, but it is a workaround. It means each user's redirected folder does not get the locked-down per-user ACL. In a client environment, that is a privacy finding waiting to be written up.

Finding 6 — `gpupdate /force` hung both clients

What happened: Drive map actions were changed to **Replace** to clear stale mappings. Both clients then froze during `gpupdate /force` and had to be power-cycled.

Actual cause: The change caused the hang. **Replace** deletes and recreates the mapping on every policy refresh. If the drive is open in Explorer or a file on it is in use, the delete blocks and the policy engine stalls.

Correction — Drive Maps action reference:

Action	Behavior	Use when
Create	Creates only if absent. Later GPO changes are ignored.	Almost never — it silently stops honoring your edits
Update	Creates if absent, modifies in place if present.	Default choice for nearly every drive map
Replace	Deletes and recreates on every refresh.	Rarely. Known cause of logon delays and <code>gpupdate</code> hangs.
Delete	Removes the mapping.	Decommissioning a share

Finding 7 — Hybrid join failed with `error_missing_device`

What happened: Users synced to Entra ID successfully, the SCP was configured, but clients reported `AzureAdJoined : NO` with `Server ErrorSubCode : error_missing_device`.

Actual cause: Hybrid join requires the **computer object** to exist in Entra ID before the device can complete registration. The OU containing the workstations was not in the Entra Connect synchronization scope, so the device object never arrived.

Correction: Install Entra Connect with **Customize**, not Express, and explicitly include both the user OU and the workstation OU in sync scope. Configure hybrid join in the same sitting. Then force a full sync rather than waiting for the 30-minute cycle:

```
Start-ADSyncSyncCycle -PolicyType Initial
```

Finding 8 — The device ownership chain (the most instructive failure in the build)

This one is worth studying closely because it is a textbook example of a shortcut causing an apparently unrelated failure days later.

Stage	What happened
Day 3	Intune auto-enrollment was not working. As a workaround, the technician signed into WS-SALES01 as <code>DOMAIN\Administrator</code> and manually added a work account through Settings. The device enrolled. Problem apparently solved.
Day 6	Microsoft 365 Apps would not deploy to WS-SALES01. Company Portal showed a "functionality limited" banner. Hours were spent chasing a missing Intune Management Extension, clearing enrollment registry keys, and attempting manual agent installs.
Resolution	WS-SALES01 was abandoned in favor of WS-ACCT01, which had been enrolled correctly via GPO and worked immediately.

Actual cause: Manually adding a work account through Settings is *user-driven enrollment*, and Intune marks devices enrolled that way as **Personal**. Personal-owned devices have deliberately reduced management capability. Devices enrolled through GPO auto-enrollment, Autopilot, or a pre-registered corporate identifier are marked **Corporate** and get full management.

The day-3 shortcut caused the day-6 failure. The two were never connected during troubleshooting.

Correction — you do not have to abandon the device:

```
Intune admin center → Devices → All devices → [select device]
→ Properties → Device ownership → change to Corporate
```

Prevention — set ownership correctly before enrollment ever happens:

```
Intune admin center → Devices → Enrollment → Corporate device identifiers
→ Add serial numbers (CSV import)
```

THE TRANSFERABLE LESSON

When a workaround gets you past a blocker, write down what you did and why. A workaround is a modification to the environment, and the environment remembers it long after you have forgotten. The most expensive troubleshooting sessions are the ones where the cause was introduced by you, days earlier, in a step you no longer remember taking.

Finding 9 — The Intune Management Extension was a red herring

What happened: Hours were spent trying to force the Intune Management Extension (IME) to install, including attempting a manual download from a link that no longer exists.

Actual cause: The IME is required for **Win32 apps, PowerShell scripts, and remediations**. It is *not* required to deploy Microsoft 365 Apps, which uses the Office CSP over the standard MDM channel. The IME's absence was a symptom of the incomplete Personal-ownership enrollment, not the cause of the app failing to install.

TROUBLESHOOTING DISCIPLINE

Before spending time on a component, confirm the component is actually in the path for what you are trying to do. Ask: *what mechanism is supposed to deliver this?* Microsoft 365 Apps → Office CSP. A packaged .intunewin app → IME. A configuration profile → OMA-DM. Different pipes, different failure modes. Chasing the wrong pipe is how an afternoon disappears.

Finding 10 — On-premises AD groups cannot target devices in Intune

What happened: The app was assigned to the on-premises `WORKSTATIONS` group. Nothing deployed. A note was made that group sync scope needed configuring.

Actual cause, and a needed correction: Intune only understands Entra ID groups — that part is right. But the deeper point is that **synchronizing on-premises groups does not solve this for devices**. Computer objects sync to Entra ID as *device* objects, and an on-premises group's membership of computer objects does not translate into Entra device group membership. You cannot fix device targeting by widening group sync scope.

Correction — use Entra dynamic device groups. This is what production environments do, it requires no maintenance, and it scales to any fleet size.


```
Entra admin center → Groups → New group
Group type:      Security
Membership type: Dynamic Device
```

Rule examples:

```
All hybrid-joined devices:
(device.deviceTrustType -eq "ServerAD")

All corporate Windows workstations by naming convention:
(device.displayName -startsWith "WS-") and (device.deviceOSType -eq "Windows")

All company-owned devices:
(device.deviceOwnership -eq "Company")
```

ANSWERING THE ORIGINAL QUESTION PROPERLY

The instinct — “*recreating groups so Intune can see them would get confusing at scale, there has to be a better way*” — was correct, and the better way is dynamic membership. Define the rule once. Every device that matches joins automatically, forever, with no human step. Group sync from on-premises remains useful for **user** groups; it is the wrong tool for devices.

Finding 11 — Disabling IPv6 on a domain controller is contrary to Microsoft guidance

What happened: DC02 promotion failed with a credential error. Diagnostics showed DNS returning public IPv6 addresses for the domain, and `nltest` contacting DC01 over IPv6. The suggested fix was to unbind IPv6 from the network adapter. It did not resolve the problem.

Two corrections here.

First, the advice itself was wrong. Microsoft explicitly recommends against disabling IPv6 on domain controllers. Windows components assume IPv6 is present, and unbinding it produces obscure failures that are harder to diagnose than whatever it was meant to fix.

Second, the actual problem was upstream. DC01 had registered ISP-assigned public IPv6 addresses in AD-integrated DNS. Domain members were then trying to reach it over an address that was not reliably routable within the VLAN.

Correct fixes, in order of preference:

1. Stop the router from advertising a global IPv6 prefix on the server VLAN. This is a router-side change and it fixes the cause.
2. If IPv6 is wanted internally, use Unique Local Addresses with stable router advertisements rather than dynamic ISP prefixes that change.
3. Prevent registration of the unwanted addresses in DNS rather than disabling the protocol stack.

Finding 12 — A virtualized domain controller must not take time from its hypervisor host

What happened: During DC02 troubleshooting, `w32tm /query /status` reported `Source: VM IC Time Synchronization Provider` and `Stratum: 1`. This was assessed as “usually fine.”

It is not fine. A virtualized domain controller pulling time from the Hyper-V Time Synchronization integration service creates a competing time hierarchy. The VM believes it is an authoritative stratum-1 source while Active Directory expects it to take time from the domain hierarchy rooted at the PDC emulator. The result is intermittent, hard-to-diagnose Kerberos failures — exactly the class of problem being troubleshot at the time.

Correction — do this on every virtualized domain controller:

```
# On the Hyper-V host
Get-VMIntegrationService -VMName "DC02" -Name "Time Synchronization" | Disable-VMIntegrationService

# On the DC itself, after promotion – take time from the domain hierarchy
w32tm /config /syncfromflags:domhier /update
Restart-Service w32time
w32tm /resync
w32tm /query /source      # should NOT say "VM IC Time Synchronization Provider"
```

Finding 13 — The failover test did not prove failover

This is the most important correction in this document, because it is the one most likely to be repeated in an interview as an accomplishment.

What happened: DC01 was powered off. A user successfully logged into a client workstation. This was recorded as successful failover.

What actually happened: The login almost certainly succeeded on **cached credentials**. Windows caches the last several domain logons locally and will authenticate a previously-seen user with no domain controller reachable at all.

Why real failover was not possible: The clients had their DNS server set **statically to DC01's IP address only**. With DC01 off, those clients could not resolve the `_ldap._tcp` SRV records that locate a domain controller. They could not have found DC02 even though DC02 was running and healthy.

The supporting evidence is in the same session: Group Policy Management on DC02 failed to open while DC01 was down, and had to be manually pointed at DC02. That is the environment telling you it still assumes DC01 is the only DC.

Corrections required to make redundancy real:

Component	Current state	Required state
Client DNS	Static, DC01 only	Delivered by DHCP, listing both DCs
DC01 DNS client settings	Itself	Primary: DC02 · Secondary: itself (127.0.0.1)
DC02 DNS client settings	DC01	Primary: DC01 · Secondary: itself (127.0.0.1)
DHCP service	Router or DC01 only	Split scope or failover across both DCs, or reservation-aware router config
FSMO roles	All five on DC01	Documented, with a seizure procedure written (SOP-07)
DC02 hosting	VM on the primary desktop workstation	Independent always-on hardware

A SECOND DC ON A MACHINE THAT GETS TURNED OFF IS NOT REDUNDANCY

DC02 lives on a desktop PC that is powered down when not in use. A domain controller that is offline half the time is worse than no second DC: it accumulates replication lag, and it can serve stale directory data when it does come back. Either give it always-on hardware or shut it down properly and demote it when the lab is idle.

Finding 14 — How to test failover correctly

Use this procedure instead. It tests the thing you actually want to know.

- 1. Fix client DNS delivery first (both DCs, via DHCP). Without this, the test is meaningless.
- 2. On a test client, clear cached credentials for the test account, or use an account that has **never** logged into that machine. Cached credentials are what invalidate the naive test.
- 3. Power off DC01.
- 4. On the client: `nltest /dsgetdc:ad.contoso.com` — it must return **DC02**.
- 5. Log in with the never-before-used account. Success proves live authentication against DC02.
- 6. `gpresult /r` — confirm policy applied and note which DC it came from.
- 7. Open a mapped drive. If shares live only on DC01, they are gone — file services were never made redundant, only the directory was.
- 8. Power DC01 back on. Run `repadmin /replsummary` and confirm replication converges with no errors.

Finding 15 — Remote Desktop is the wrong tool for helpdesk support

What happened: RDP was enabled on client workstations via GPO to allow remote management. It worked.

The problem: Windows client operating systems support only **one interactive session**. When you RDP into a user's workstation, that user is disconnected from their own desktop. You cannot watch what they are doing, you cannot show them anything, and you have interrupted them to help them.

Tool	Use for	User stays logged in?
RMM remote control	Day-to-day helpdesk. The professional answer.	Yes — shared session
Quick Assist	Ad-hoc support, no infrastructure required	Yes — shared session
RDP to a workstation	Unattended work on an idle machine	No — kicks the user off
RDP to a server	Server administration	N/A — servers allow multiple sessions

NEVER EXPOSE RDP TO THE INTERNET

Not on 3389, not on an obscure high port. Internet-facing RDP is among the most common ransomware entry points in the SMB space. If you assess a client network and find RDP open to the world, that is your first finding and it is urgent. Acceptable paths: RMM agent, VPN then RDP, RD Gateway with MFA, or Azure Bastion.

If RDP is enabled anyway, require Network Level Authentication and scope who may connect:

```
Computer Configuration → Policies → Administrative Templates → Windows Components
→ Remote Desktop Services → Remote Desktop Session Host → Security
"Require user authentication for remote connections by using
Network Level Authentication" = Enabled
```

Finding 16 — Administration was performed from the domain controller console

What happened: Group Policy Management was missing on DC02 and had to be installed there. Nearly all administration throughout the build was performed while logged into a DC.

Why this is a bad habit: Every browser session, every downloaded installer, and every logged-in admin token on a domain controller is a security exposure on the most sensitive machine in the environment. It is also fragile — as demonstrated when DC01 went offline and the management console followed it.

Correction: Build a management workstation. Install RSAT on a normal Windows 11 machine and administer everything remotely.

```
# On a Windows 11 admin workstation, elevated
Add-WindowsCapability -Online -Name Rsat.ActiveDirectory.DS-LDS.Tools~0.0.1.0
Add-WindowsCapability -Online -Name Rsat.GroupPolicy.Management.Tools~0.0.1.0
Add-WindowsCapability -Online -Name Rsat.DHCP.Tools~0.0.1.0
Add-WindowsCapability -Online -Name Rsat.Dns.Tools~0.0.1.0
Add-WindowsCapability -Online -Name Rsat.FileServices.Tools~0.0.1.0
```

Finding 17 — Evaluation licensing has a hard deadline

Both domain controllers run Windows Server Evaluation, which expires after **180 days**. When it expires, the server begins shutting down on a schedule.

YOU CANNOT CONVERT AN EVALUATION SERVER TO LICENSED WHILE IT IS A DOMAIN CONTROLLER

Microsoft blocks in-place conversion of an evaluation installation to a retail or volume licensed edition on a machine holding the AD DS role. To convert, you must demote the DC first, convert, then re-promote. Plan for this before the timer runs out, or the lab will simply stop working with no clean path forward.

Check remaining time with: `slmgr /dlv`

Finding 18 — Diagnostic commands were run at a stage where they could not work

What happened: `Test-ComputerSecureChannel` was suggested on DC02 before DC02 had joined the domain. It returned *"The local computer is not currently part of a domain"* — the only answer it could have given.

Lesson: Before running a diagnostic, know its precondition. Time spent interpreting the output of an inapplicable test is worse than wasted — it introduces noise into the picture you are building. When a diagnostic returns something confusing, first ask whether it was valid to run at all.

Finding 19 — An empty licensing group breaks Intune, silently

What happened (Round 2): Group-based licensing was configured — a `LIC-M365-BusinessPremium` group was created and the SKU assigned to it. Users synced correctly. Devices hybrid joined correctly. Intune enrollment then failed with no error anywhere.

Actual cause: nobody was ever added to the licensing group. The group existed, the SKU was attached, and it licensed zero users.

Why it is hard to trace: the failure surfaces two phases later and in a different product. The MDM auto-enrollment GPO runs in the *user's* context, so an unlicensed user cannot enroll a device — but Intune does not report "unlicensed user," it simply never shows the device. Working backwards from an absent device to an empty group is a long walk.

Correction: nest the synced `GG-AllEmployees` group into the cloud licensing group, once. Every future user inherits a license automatically through the same chain that gives them file access. Full procedure in Phase 9.6.

ON DIRECT LICENSE ASSIGNMENT

Assigning a license straight to a user is the correct *diagnostic* step — it isolates whether the problem is the license or the group plumbing. That is not a contradiction of the group model. Remove the direct assignment once the root cause is fixed, or you create a user whose licensing survives removal from every group, which is exactly the offboarding gap group licensing exists to prevent.

Finding 20 — Read the error text before rewriting the script

What happened (Round 2): The share creation block threw errors. They were diagnosed as a missing parameter, and the script was modified to add it.

Actual cause: the error read *"The name has already been shared"* — Windows System Error 2118. The shares had been created by a previous partial run. `New-SmbShare` only creates; it will not update an existing share. The script was correct.

The transferable lesson: when a command fails, read what it actually says before forming a theory. Rewriting correct code because of a misread error costs time twice — once to make the change, and again later when the same error reappears and the earlier "fix" is now confusing evidence.

Correction: make setup scripts idempotent so a partial run can be safely repeated. Phase 7.2 now checks for existence and calls `Set-SmbShare` instead of failing.

Finding 21 — A DHCP pool spanning the whole subnet

What happened (Round 2): The firewall's DHCP scope was left at its default of `.1` through `.254` — the entire subnet — while the gateway, DC01, and DC02 held static addresses inside that range.

Why it had not broken yet: with only a handful of devices, DHCP had not yet reached those addresses. It is a latent fault, not a dormant one.

What it looks like when it fires: a client is handed the domain controller's address. Authentication, DNS, and file services fail simultaneously across the whole site, and nothing on the server indicates why. This is a genuinely nasty outage to diagnose under pressure.

Correction: reserve the bottom of every range for infrastructure — start the pool at `.100`, or configure an explicit exclusion range. Phase 4.

Finding 22 — Entra roles do not grant Azure permissions

What happened (Round 2): A daily-driver admin account was created and assigned Global Administrator and Intune Administrator in Entra ID. It then had no ability to view or modify the Azure subscription at all.

Actual cause: Entra ID roles and Azure RBAC are separate authorization systems. Entra roles govern the *directory* — users, groups, devices, tenant configuration. Azure RBAC governs *resources* — subscriptions, resource groups, virtual machines. Global Administrator confers no resource access by design.

Correction, and it was handled correctly at the time: sign in with the account that owns the subscription, assign **Contributor** at subscription scope to the working admin account, and retain Owner on the break-glass account. Preferring Contributor over Owner for daily work is the right instinct — Owner can also grant access to others, which is rarely needed.

Finding 23 — Time zone is a Kerberos dependency

What happened (Round 2): A laptop imaged in one time zone was deployed in another. Domain join failed with "An Active Directory Domain Controller could not be contacted." DNS was correct and the DC pinged fine.

Actual cause: one hour of clock skew. Kerberos tolerates five minutes. The KDC rejected the authentication request outright, and the client — unable to complete authentication — reported it as an inability to contact a domain controller.

Why the error misleads: the message describes the symptom, not the cause. The DC was contacted. It said no.

Correction: set the time zone before attempting any join, and add the skew check to the pre-join block:

```
w32tm /stripchart /computer:<DC-IP> /samples:3 /dataonly
```

Five seconds, and it eliminates an entire category of failure before you touch DNS or firewall rules.

Summary — errors, causes, and correct fixes

Symptom	Actual root cause	Correct fix
Domain join fails, domain not found	Client DNS not pointing at a DC	Fix DHCP option 6; never set client DNS statically
GPO absent from <code>gpresult</code>	Linked to an OU that does not contain the object	Link where the user or computer object lives
New PC receives no policy	Object landed in <code>CN=Computers</code>	<code>redircmp</code> to a real OU
Can browse share, cannot open files	NTFS permissions	Grant Modify to the resource group; check "Applies to"
Total access loss after cleanup	<code>SYSTEM</code> / admin removed from ACL	Take ownership, rebuild ACL from the standard model
<code>gpupdate /force</code> hangs	Drive map action = Replace, drive in use	Change action to Update
Folder Redirection Event 502	Home and redirected roots collide; folder not owned by user	Separate the roots
<code>AzureAdJoined : NO</code>	Computer OU not in Entra Connect sync scope	Add OU to scope, full sync, <code>dsregcmd /leave</code> then <code>/join</code>
Intune apps never install	Device ownership = Personal from manual enrollment	Change ownership to Corporate; enroll via GPO
Intune assignment does nothing	Targeted an on-prem group that cannot contain Entra devices	Use an Entra dynamic device group
Second DC promotion, bad credentials	Name resolution / IPv6 registration; time hierarchy	Join as member server first, fix time source, then promote
Arc agent not connected	Expired onboarding token	<code>azcmagent disconnect --force</code> then <code>connect</code>
<code>azcmagent</code> not recognized	PATH not refreshed in current session	New elevated session, or call by full path
Multi-line PowerShell fails oddly	Whitespace after a backtick	Collapse to a single line
GPMC fails when a DC is down	Console targeting the offline DC	Change Domain Controller; fix DNS redundancy properly
VM cannot reach the LAN	Attached to Hyper-V Default Switch (NAT)	Attach to an External virtual switch

Part 3 — The Build Playbook

This is the greenfield build in correct order. Every phase assumes the previous one is complete and verified.

Estimated time on a competent second pass: Phases 0–9 in one working day, Phases 10–14 in a second. The first time will take three times that, and that is normal.

Build order at a glance

Phase	What	Why it is here and not later
0	Design decisions	Naming and addressing are expensive to change afterward
1	Server base build	Rename and set IP before promotion; both are painful after
2	Promote first DC, DNS, time	Time and DNS underpin every authentication that follows
3	Backup and AD Recycle Bin	Before there is anything to lose. Recycle Bin must be on before a deletion, not after.
4	DHCP and DNS delivery	Clients need correct DNS before they can join anything
5	OU structure and container redirection	Must exist before any object is created
6	Groups	Permissions and policy filtering both depend on them
7	File services	Shares must be permissioned before drive maps reference them
8	Group Policy	Built and tested before real users exist
9	Cloud identity and UPN suffix	Before creating users — otherwise every UPN gets rewritten
10	Create users	Now every account is correct at birth
11	Client build and join	The first user logs into a finished environment
12	Intune policy and apps	Devices are already enrolled by the Phase 8 GPO
13	Second DC and real redundancy	Once the single-DC build is proven correct
14	Operations and handover	The part that keeps the client

Phase 0 — Design decisions

Do this on paper before touching hardware. Every item here is expensive to change later.

0.1 Domain name

Client owns	Use	Never use
contoso.com	ad.contoso.com or corp.contoso.com	contoso.local
Nothing yet	Register a domain first — roughly \$12 per year	contoso.local

WHY

.local is reserved for multicast DNS, can never be verified in Entra ID, and cannot receive a public certificate. Choosing a routable subdomain now eliminates the UPN rewrite in Phase 9 entirely and keeps every future cloud option open. See Part 2, Finding 1.

0.2 Naming conventions

Object type	Pattern	Example
Domain controller	<SITE>-DC##	RC-DC01
Member server	<SITE>-<ROLE>##	RC-FILE01 , RC-SYNC01
Workstation	WS-<DEPT>##	WS-SALES01
Role group (Global)	GG-<Name>	GG-Sales
Resource group (Domain Local)	DL-<Resource>-<Access>	DL-Sales-Modify
Service account	svc-<purpose>	svc-backup
GPO	<Scope> - <Purpose>	Workstations - Baseline Security
Entra dynamic group	DYN-<Scope>	DYN-Windows-Corporate

0.3 Addressing plan

Write down and keep: server subnet and VLAN, the static range reserved for infrastructure (typically .1 to .50), the DHCP scope range, the gateway, the DNS servers to be handed out, and the planned IP of every server.

0.4 Cloud tenant

Create the tenant during design, not during Phase 9. Tenant names are globally unique and first-come. The name you want may already be taken, and you want to discover that before anything is built around it.

0.5 Licensing

Requirement	What it needs
Intune device management	Business Premium, E3, E5, or standalone Intune
Group-based licensing	Entra ID P1 (included in Business Premium and E3)
Conditional Access	Entra ID P1
Azure Arc, Log Analytics, Azure Backup	A real Azure subscription — a Microsoft 365 subscription is not one
Windows Server in production	A licensed edition. Evaluation expires at 180 days and cannot be converted while the server is a domain controller.

0.6 Credential handling

Create the vault entry before creating the first password. It must contain the local administrator password, the DSRM password, the domain administrator account, a dedicated break-glass account, and the tenant global administrator.

WHY

The DSRM password is needed exactly once — during directory recovery — and it is reliably the one nobody wrote down. The break-glass account exists so that a Conditional Access misconfiguration cannot lock every administrator out of the tenant at once. Make it cloud-only, exclude it from all CA policies, secure it with a hardware key, and alert on its use.

Phase 1 — Server base build

Perform in this exact order. Steps 5 and 6 are painful to reverse after promotion.

1. Install Windows Server, Standard edition, Desktop Experience.
2. Set a strong local Administrator password. Record it in the vault immediately.
3. Patch fully. Reboot. Patch again until no updates remain.
4. Set the correct time zone.
5. **Rename the server** to its final hostname. Reboot.
6. **Set the static IP**: address, mask and gateway per the Phase 0.3 plan; Preferred DNS = **the server's own static IP**; Alternate DNS = 127.0.0.1 .
7. Configure storage: operating system on one volume, data on a separate volume.
8. Install vendor drivers and update firmware.

WHY RENAME BEFORE PROMOTION

Renaming a member server is a reboot. Renaming a domain controller requires `netdom computername` plus service principal name cleanup and carries real risk. Two minutes of forethought avoids an hour of repair.

VERIFY

```
Get-ComputerInfo | Select-Object CsName, OsName, WindowsVersion
Get-NetIPConfiguration
Get-Volume
```

Phase 2 — Promote the first domain controller

2.1 Promotion

1. Server Manager, Add Roles and Features, **Active Directory Domain Services**. DNS installs with it.
2. Click the notification flag, then **Promote this server to a domain controller**.
3. **Add a new forest**, root domain name from Phase 0.1.
4. Forest and domain functional level: highest available.
5. Check **DNS server** and **Global Catalog**.
6. Set the DSRM password. **Record it in the vault now**.
7. The DNS delegation warning is expected for a new forest. Continue.
8. Reboot.

2.2 DNS hygiene, immediately after reboot

1. **Forwarders**. DNS Manager, server Properties, Forwarders. Add upstream resolvers such as `1.1.1.1` and `8.8.8.8`, or the client's filtering service.
2. **Reverse lookup zone**. Create one for the server subnet. Two minutes, and it is what makes `nslookup <ip>` work when you are troubleshooting under pressure.
3. **Scavenging**. DNS server Properties, Advanced, Enable automatic scavenging, 7 days. Set aging on the forward zone.

CHECK IPV6 REGISTRATION — AND CLEAN UP PROPERLY

Run `Resolve-DnsName ad.contoso.com`. If public IPv6 addresses appear (typically ISP-assigned `2xxx:` globals), fix it at the router by not advertising a global IPv6 prefix on the server VLAN. **Do not unbind IPv6 from the adapter** — Microsoft advises against that on domain controllers. See Part 2, Finding 11.

Turning off router advertisements is only step one. Addresses already assigned to the NIC persist until the interface drops them, and AAAA records already written to the zone persist until they are removed or scavenged. Restarting the adapter is not enough. Complete the cleanup:

```
# 1. Confirm what the DC actually holds now
Get-NetIPAddress -AddressFamily IPv6 | Select-Object InterfaceAlias, IPAddress, PrefixOrigin, SuffixOrigin

# 2. Reboot after disabling router advertisements – this drops SLAAC-assigned addresses

# 3. Delete the stale AAAA records from the zone
Get-DnsServerResourceRecord -ZoneName "ad.contoso.com" -RRType AAAA |
  Remove-DnsServerResourceRecord -ZoneName "ad.contoso.com" -Force

# 4. Force the DC to re-register
ipconfig /registerdns
Restart-Service netlogon

# 5. Verify – only the A record should remain
Resolve-DnsName ad.contoso.com
```

Leaving stale AAAA records in place is what produced the misleading DC02 promotion failure in round 1: clients resolved the domain to an unroutable IPv6 address and reported a credential error.

2.3 Time — configure now, not later

On the PDC emulator, which is the first DC by default:

```
w32tm /config /manualpeerlist:"time.windows.com,0x9 pool.ntp.org,0x9" /syncfromflags:manual /reliable:yes /update
Restart-Service w32time
w32tm /resync
w32tm /query /status
```


WHY TIME IS A FOUNDATION ITEM

Kerberos rejects tickets with more than five minutes of clock skew. Every domain member slaves its clock to the PDC emulator. If the PDC drifts, every authentication in the environment fails at once — and it presents as "everything is broken," not as a clock problem. Setting the external anchor on day one prevents an entire category of future outage.

VERIFY

```
dcdiag /v                # expect all tests passing
nslookup ad.contoso.com  # returns the DC's IP
netdom query fsmo         # all five roles listed
w32tm /query /source      # an external NTP server, not "Local CMOS Clock"
Get-DnsServerForwarder
```

Phase 3 — Backup and AD Recycle Bin

Do this before creating a single object.

3.1 AD Recycle Bin

```
Enable-ADOptionalFeature 'Recycle Bin Feature' -Scope ForestOrConfigurationSet -Target 'ad.contoso.com'
```

ENABLE IT BEFORE YOU NEED IT

The Recycle Bin only protects objects deleted *after* it is enabled, and it cannot be turned off once on. It converts "restore the whole directory from backup" into a two-minute undelete. There is no reason not to enable it on day one, and almost every environment you inherit will not have it.

3.2 System State backup

```
Install-WindowsFeature Windows-Server-Backup

# One-off backup to an external or network target
wbadmin start systemstatebackup -backupTarget:E: -quiet

# Scheduled daily
$Policy = New-WBPolicy
Add-WBSystemState -Policy $Policy
$Target = New-WBBackupTarget -VolumePath "E:"
Add-WBBackupTarget -Policy $Policy -Target $Target
Set-WBSchedule -Policy $Policy -Schedule 22:00
Set-WBPolicy -Policy $Policy -Force
```

VERIFY

```
Get-WBSummary
Get-ADOptionalFeature -Filter 'name -like "Recycle Bin Feature"'
```

Then test a restore. A backup you have never restored from is a hypothesis, not a backup.

Phase 4 — DHCP and client DNS delivery

Decide who serves DHCP. Prefer Windows DHCP when a server exists, because it integrates with DNS and is centrally auditable.

```
Install-WindowsFeature DHCP -IncludeManagementTools
Add-DhcpServerInDC

Add-DhcpServerv4Scope -Name "LAN" -StartRange 10.10.10.100 -EndRange 10.10.10.200 -SubnetMask 255.255.255.0

# Hand out BOTH domain controllers as DNS, in order
Set-DhcpServerv4OptionValue -ScopeId 10.10.10.0 -DnsServer 10.10.10.10,10.10.10.11 -DnsDomain "ad.contoso.com" -Router 10.10.10.1
```

If the router or firewall serves DHCP instead, set its DNS option for that VLAN to both domain controllers.

THE DHCP POOL MUST NOT OVERLAP YOUR STATIC ADDRESSES

This is the most commonly missed step on consumer and prosumer firewalls, which frequently default the pool to the entire subnet — .1 through .254 . If your gateway is .1 and your domain controllers are .10 and .20 , those addresses sit inside the pool and DHCP is free to lease them to a client.

The failure is intermittent and vicious: everything works until the day a laptop is handed the DC's address. Then authentication, DNS, and file services fail simultaneously across the whole site, and the cause is invisible from the server.

Reserve the bottom of the range for infrastructure. Start the DHCP pool at .100 , or configure an explicit exclusion:

```
# Windows DHCP – exclude the static infrastructure range
Add-DhcpServerv4ExclusionRange -ScopeId 10.10.10.0 -StartRange 10.10.10.1 -EndRange 10.10.10.99
Get-DhcpServerv4ExclusionRange -ScopeId 10.10.10.0
```

On a firewall appliance, set the pool's start address above your static block instead. Also check for a separate **Search Domain** or **Domain Name** field alongside the DHCP options — some appliances ship it defaulted to `lan` or `local` , which will conflict with the DNS suffix you set in option 15. Set both to your AD domain.

NEVER SET CLIENT DNS STATICALLY

Static client DNS is exactly how the lab ended up with no working failover: every client pointed at one DC by hand, so when that DC went down nothing could locate the second one. DHCP option 6 is the correct delivery mechanism. It scales, it survives an IP change, and it is what makes adding a second DC actually mean something.

VERIFY FROM A CLIENT, NOT FROM THE SERVER

```
ipconfig /all           # both DCs listed as DNS servers
nslookup ad.contoso.com
nltest /dsgetdc:ad.contoso.com
```

Phase 5 — OU structure

5.1 Build the tree

```
$Root = "DC=ad,DC=contoso,DC=com"
New-ADOrganizationalUnit -Name "Contoso" -Path $Root -ProtectedFromAccidentalDeletion $true

$Org = "OU=Contoso,$Root"
"Employees","Workstations","Servers","Groups","ServiceAccounts","Disabled" | ForEach-Object {
    New-ADOrganizationalUnit -Name $_ -Path $Org -ProtectedFromAccidentalDeletion $true
}
```

DO NOT NAME TOP-LEVEL OUS "USERS" OR "COMPUTERS"

Those names collide with the built-in containers `CN=Users` and `CN=Computers` . Beyond the collision, those containers cannot hold OUs and **cannot have a GPO linked to them** — anything sitting in them receives only the Default Domain Policy. The single-organization-OU pattern above avoids the problem entirely.

5.2 Redirect the default containers

```
redircmp "OU=Workstations,OU=Contoso,DC=ad,DC=contoso,DC=com"
redirusr "OU=Employees,OU=Contoso,DC=ad,DC=contoso,DC=com"
```

WHY THIS ONE COMMAND MATTERS SO MUCH

By default every machine that joins the domain lands in `CN=Computers` , where no GPO reaches it. That is the number one cause of "the new PC didn't get its settings." `redircmp` changes the default landing spot to a real OU, so every future machine receives the workstation baseline automatically with no manual step. It converts a checklist item everyone forgets into something that cannot be forgotten.

VERIFY

```
Get-ADOrganizationalUnit -Filter * | Select-Object Name, DistinguishedName
Get-ADDomain | Select-Object ComputersContainer, UsersContainer
```

Phase 6 — Security groups

Build both tiers of the AGDLP model before any user or any permission exists.

```
$OU = "OU=Groups,OU=Contoso,DC=ad,DC=contoso,DC=com"

# Role groups - describe WHO SOMEONE IS
"GG-Sales","GG-Accounting","GG-IT","GG-AllEmployees" | ForEach-Object {
    New-ADGroup -Name $_ -GroupScope Global -GroupCategory Security -Path $OU
}
Add-ADGroupMember -Identity "GG-AllEmployees" -Members "GG-Sales","GG-Accounting","GG-IT"

# Resource groups - describe WHAT ACCESS IS GRANTED
"DL-Company-Modify","DL-Sales-Modify","DL-Sales-Read","DL-Accounting-Modify","DL-IT-Modify" | ForEach-Object {
    New-ADGroup -Name $_ -GroupScope DomainLocal -GroupCategory Security -Path $OU
}

# Nest role groups into resource groups
Add-ADGroupMember -Identity "DL-Company-Modify" -Members "GG-AllEmployees"
Add-ADGroupMember -Identity "DL-Sales-Modify" -Members "GG-Sales"
Add-ADGroupMember -Identity "DL-Accounting-Modify" -Members "GG-Accounting"
Add-ADGroupMember -Identity "DL-IT-Modify" -Members "GG-IT"
```

AGDLP, AND WHY TWO TIERS

Accounts into **Global** groups, Global groups into **Domain Local** groups, Permissions onto the Domain Local group.

In a single domain you can put the Global group straight onto the ACL and it works. The separation buys you this: when someone asks why Sales can write to a folder, the answer is one entry on that folder's ACL. When auditors need read access for a month, you add them to [DL-Sales-Read](#) and remove them later — you never touch the folder again. Changing group membership is instant and reversible. Re-permissioning a folder tree is neither.

NEVER ASSIGN PERMISSIONS TO AN INDIVIDUAL USER ACCOUNT

Not once, not temporarily. Direct user entries on an ACL are invisible to any group-based audit, and they are why departed employees keep access to things nobody can explain.

Phase 7 — File services

7.1 Folder layout

```
D:\
├── Company\           company-wide files
├── Departments\
│   ├── Sales\
│   ├── Accounting\
│   └── IT\
├── Home\             H: home drives
└── Redirected\       Folder Redirection target – SEPARATE from Home
```

HOME AND REDIRECTED MUST BE DIFFERENT ROOTS

Pointing AD home folders and Folder Redirection at the same path causes Event ID 502 "Access is denied" that survives every permission fix you will try. AD's home-folder feature creates the folder owned by the admin who clicked Apply; Folder Redirection with exclusive rights then refuses because the user does not own it. Separate roots make the conflict impossible. See Part 2, Finding 5.

7.2 Create the shares

Use the idempotent version. It can be re-run safely, which matters because the most common failure here is re-running the script after a partial success.

```
$Shares = @(
    @{Name="Company";      Path="D:\Company"},
    @{Name="Departments"; Path="D:\Departments"},
    @{Name="Home";         Path="D:\Home"},
    @{Name="Redirected";   Path="D:\Redirected"}
)

foreach ($S in $Shares) {
    if (-not (Test-Path $S.Path)) { New-Item -Path $S.Path -ItemType Directory -Force | Out-Null }

    if (Get-SmbShare -Name $S.Name -ErrorAction SilentlyContinue) {
        Write-Warning "$($S.Name) already shared - updating instead of creating"
        Set-SmbShare -Name $S.Name -FolderEnumerationMode AccessBased -Force
    }
    else {
        New-SmbShare -Name $S.Name -Path $S.Path `
            -FullAccess "Authenticated Users" `
            -FolderEnumerationMode AccessBased
        Write-Host "Created share $($S.Name)" -ForegroundColor Green
    }
}

Get-SmbShare | Where-Object { $_.Name -notlike "*$" } |
Select-Object Name, Path, FolderEnumerationMode
```

"THE NAME HAS ALREADY BEEN SHARED" — WINDOWS SYSTEM ERROR 2118

`New-SmbShare` creates only. If the share name already exists it throws error 2118 and does nothing — it will not update the existing share. Partially completing this step and then re-running the whole block produces a wall of these errors while the shares that *did* succeed sit there working fine. The error text names the cause precisely. Read it before assuming the script is wrong — misattributing 2118 to a syntax or parameter problem sends you rewriting a script that was correct. To change an existing share, use `Set-SmbShare`. To start over, `Remove-SmbShare -Name X -Force` then re-create.

WHY THE SHARE LAYER IS LEFT WIDE OPEN

Share and NTFS permissions are evaluated together and the more restrictive wins. Enforcing security at both layers means every access problem has two candidate causes, and you will troubleshoot the wrong one about half the time. The industry standard is to open the share to Authenticated Users and let NTFS carry all the real control. One place to look, one place to fix.

7.3 Access-Based Enumeration

`-FolderEnumerationMode AccessBased` enables ABE. Users do not see folders they lack Read access to — the Accounting folder is invisible to Sales rather than visible-but-locked.

```
Set-SmbShare -Name "Departments" -FolderEnumerationMode AccessBased -Force
```

ABE VERSUS ITEM-LEVEL TARGETING

These solve different halves of the same problem. Item-level targeting controls whether a **drive letter** appears. ABE controls whether a **folder** appears. Without ABE, a user who types the UNC path directly sees every department folder regardless of drive mappings. Use both.

7.4 A note on UNC paths: server name vs DFS namespace

Everything in this playbook uses `\\SERVER\Share`. That is correct, it works, and it is the right starting point.

The production upgrade is a **domain-based DFS Namespace**, which gives a server-agnostic path:

```
\\ad.contoso.com\shares\Departments\Sales      instead of      \\FILE01\Departments\Sales
```

A DFS PATH IS NOT AUTOMATIC — THE NAMESPACE MUST EXIST FIRST

`\\ad.contoso.com\shares\` does not resolve simply because it is your domain name. You must install the DFS Namespaces role and create the namespace with folder targets pointing at the real shares. Until you do, that path returns "network path was not found."

```
Install-WindowsFeature FS-DFS-Namespaces -IncludeManagementTools
New-DfsnRoot -TargetPath "\\FILE01\shares" -Type DomainV2 -Path "\\ad.contoso.com\shares"
New-DfsnFolder -Path "\\ad.contoso.com\shares\Departments" -TargetPath "\\FILE01\Departments"
```

Decide once, before you author the drive-map GPO. Switching later means editing every preference item. If you are not setting up DFS-N on this build, use the server name and move on — it is a valid choice, not a compromise. Revisit it when you have two file servers, or when a server rename or replacement is on the horizon.

7.5 NTFS permissions on a department folder

Principal	Access	Applies to
Domain Admins	Full Control	This folder, subfolders and files
SYSTEM	Full Control	This folder, subfolders and files
DL-Sales-Modify	Modify	This folder, subfolders and files
CREATOR OWNER	Full Control	Subfolders and files only

Break inheritance using **Convert inherited permissions into explicit permissions**, then remove what does not belong.

7.6 Home and Redirected root permissions — exact values

Principal	Access	Applies to
Domain Admins	Full Control	This folder, subfolders and files
SYSTEM	Full Control	This folder, subfolders and files
CREATOR OWNER	Full Control	Subfolders and files only
Authenticated Users	Traverse folder/execute file, List folder/read data, Read attributes, Read extended attributes, Create folders/append data, Read permissions	This folder only

INCLUDE TRAVERSE FOLDER/EXECUTE FILE

Omitting it usually still works, because the *Bypass traverse checking* user right is granted to Everyone by default and makes the traverse check unnecessary. But that default can be hardened away by a security baseline, and when it is, users suddenly cannot reach their own home folders. Microsoft's documented home-folder ACL includes traverse. Include it and remove the dependency on a default you do not control.

WHY THE "THIS FOLDER ONLY" SCOPING IS CRITICAL

Users need to create their own folder at the root but must not inherit rights into anyone else's. **CREATOR OWNER** then gives each user full control over the folder they personally created. This exact combination is what makes per-user private folders work without you touching each one by hand.

VERIFY WITH A THROWAWAY TEST ACCOUNT, FROM A CLIENT

- Can create and edit files in your own department folder
- Cannot write to another department's folder
- The other department's folder is **not visible at all** when browsing `\\SERVER\Departments`

```
Get-SmbShare | Select-Object Name, Path, FolderEnumerationMode
Get-Acl D:\Departments\Sales | Format-List
```

Phase 8 — Group Policy

Build every GPO now. Test with **one** throwaway user and **one** throwaway machine. Then onboard everyone else into a working environment.

8.1 Ground rules

- **Default Domain Policy:** password and account lockout policy only. Those must live at the domain root to be honored for domain accounts. Nothing else goes in it.

- **Default Domain Controllers Policy:** leave alone entirely.
- One GPO, one purpose. A forty-setting monolith cannot be troubleshooted.
- Computer Configuration links to the OU holding the *computers*. User Configuration links to the OU holding the *users*.

THE RULE THAT TRIPS EVERYONE

Group Policy applies based on where the **user or computer object** lives in the OU tree. It does not apply to groups, and it does not care where groups live. Linking a GPO to an OU full of security groups does nothing at all. Groups *filter* policy — through security filtering or item-level targeting — they never scope it.

8.2 GPOs to build

A. Default Domain Policy — passwords and logout

```
Computer Configuration > Policies > Windows Settings > Security Settings
> Account Policies > Password Policy
    Minimum password length: 14
    Password must meet complexity requirements: Enabled
> Account Lockout Policy
    Account lockout threshold: 10
    Reset counter after: 15 minutes
```

B. Workstations - Baseline Security, linked to OU=Workstations

```
Computer Configuration > Policies > Windows Settings > Security Settings
> Local Policies > Security Options
    Interactive logon: Message title for users attempting to log on
    Interactive logon: Message text for users attempting to log on
    Accounts: Guest account status = Disabled
    Interactive logon: Don't display last signed-in = Enabled

Computer Configuration > Policies > Administrative Templates > System > Logon
    Always wait for the network at computer startup and logon = Enabled
```

"ALWAYS WAIT FOR THE NETWORK"

Windows uses fast logon optimization by default, processing policy in the background from a cache. Drive maps, Folder Redirection, and software installation all need the network available *during* logon. Without this setting they can take two or three reboots to appear. When a user says "the drive shows up eventually," this is why.

C. Workstations - LAPS, linked to OU=Workstations

```
# One-time schema extension, run as a Schema Admin
Update-LapsADSchema

# Grant computers permission to write their own password
Set-LapsADComputerSelfPermission -Identity "OU=Workstations,OU=Contoso,DC=ad,DC=contoso,DC=com"

# GPO settings
Computer Configuration > Policies > Administrative Templates > System > LAPS
    Configure password backup directory = Active Directory
    Password Settings = complexity all four character sets, length 14+, age 30 days
    Enable password encryption = Enabled
```

LAPS IS NOT OPTIONAL

Without it every workstation shares one local administrator password. One compromised machine hands an attacker local administrator on every machine in the environment — this is the standard lateral movement path in ransomware incidents. Windows LAPS is built into current Windows at no cost. Its absence is a finding on any security assessment.

D. Workstations - MDM Auto-Enrollment, linked to OU=Workstations

```
Computer Configuration > Policies > Administrative Templates > Windows Components > MDM
    Enable automatic MDM enrollment using default Azure AD credentials = Enabled
    Select Credential Type to Use = User Credential
```

Build this now even though the tenant does not exist yet. It does nothing until hybrid join is configured, and then it simply works.

WHY THIS SPECIFIC GPO SAVES YOU DAYS

This is the method that produces **Corporate** device ownership and full management capability. Manually adding a work account through Settings produces **Personal** ownership and a crippled device. Having this GPO in place before the first machine joins means you never take the manual shortcut, and never inherit its consequences. See Part 2, Finding 8.

E. Employees - Drive Maps, linked to OU=Employees

User Configuration > Preferences > Windows Settings > Drive Maps

Setting	Value
Action	Update — never Replace
Location	\\SERVER\Company
Drive letter	S
Reconnect	Checked
Label as	Company
Common tab	Run in logged-on user's security context, checked
Common tab	Item-level targeting, checked, Security Group = GG-AllEmployees

Repeat one item per department, targeting the matching GG- group. Use drive letters from the end of the alphabet. Avoid A: and B: — legacy floppy letters that some line-of-business software still treats as removable media.

F. Employees - Folder Redirection, linked to OU=Employees

User Configuration > Policies > Windows Settings > Folder Redirection
For Documents, Desktop, Pictures:
Setting: Basic - Redirect everyone's folder to the same location
Target: Create a folder for each user under the root path
Root path: \\SERVER\Redirected <-- NOT the Home root
Policy Removal: Leave the folder in the new location
Grant the user exclusive rights: Enabled (safe now that roots are separate)

FOLDER REDIRECTION APPLIES AT LOGON ONLY

gpupdate /force will never show you the result. You must fully log off and log back on. Locking and unlocking is not a logoff.

G. Workstations - Remote Access, linked to OU=Workstations (optional)

Computer Configuration > Policies > Administrative Templates > Windows Components
> Remote Desktop Services > Remote Desktop Session Host
> Connections
Allow users to connect remotely by using Remote Desktop Services = Enabled
> Security
Require user authentication for remote connections by using
Network Level Authentication = Enabled

Computer Configuration > Preferences > Control Panel Settings > Local Users and Groups
Action: Update, Group: Remote Desktop Users
Add: GG-IT (scope who may connect - do not leave it open)

RDP IS FOR UNATTENDED MACHINES, NOT HELPDESK

Windows client operating systems allow one interactive session. RDP into a user's workstation and you disconnect that user. For support, use an RMM remote control tool or Quick Assist, which share the session. See Part 2, Finding 15.

8.3 Test with one user and one machine

VERIFY BEFORE PROCEEDING TO PHASE 9

```
gpresult /r # user and computer GPOs applied
gpresult /h C:\gpo.html; start C:\gpo.html # far more readable
whoami /groups # group membership resolved
net use # drive maps present
dsregcmd /status # device state
```

Confirm visually: legal banner at logon, drives mapped, Documents Properties Location shows the server path, no Folder Redirection errors in Event Viewer Application log.

Fix everything here. Every defect you leave becomes a per-machine repair after Phase 11.

Phase 9 — Cloud identity, before creating users

This phase sits here for one reason: the UPN suffix is a property of every user account. Setting it correctly at creation costs nothing. Fixing it afterward means rewriting every account in the directory.

9.1 Tenant and custom domain

1. Start the Microsoft 365 subscription or trial. Create the global administrator. Record in the vault.
2. Microsoft 365 admin center, Settings, Domains, Add domain, [contoso.com](#), verify via the TXT record.
3. Create a cloud-only **break-glass** global administrator on the [.onmicrosoft.com](#) domain. Exclude it from all Conditional Access. Secure with a hardware key.

9.2 Add the UPN suffix in Active Directory

```
Active Directory Domains and Trusts > right-click the top node > Properties
> UPN Suffixes tab > add: contoso.com > Apply
```

If you inherited a [.local](#) domain, this is also the remediation — add the routable suffix and rewrite existing UPNs:

```
Get-ADUser -Filter * -SearchBase "OU=Employees,OU=Contoso,DC=ad,DC=contoso,DC=com" |
ForEach-Object { Set-ADUser $_ -UserPrincipalName "$($_.SamAccountName)@contoso.com" }
```

9.3 Install Entra Connect

INSTALL ON A MEMBER SERVER, NOT A DOMAIN CONTROLLER

Entra Connect runs a SQL Express instance and a service account with directory-wide read access. On a DC that expands the attack surface of the most sensitive machine you have and complicates both patching and DC recovery. Microsoft's documented guidance is a dedicated member server. In a single-box lab you will install it on the DC and it will work — know that it is a lab compromise and say so if asked.

WHICH PRODUCT: CONNECT SYNC OR CLOUD SYNC?

Microsoft ships two sync tools and the choice is not a preference — it is determined by whether you need hybrid join.

Capability	Entra Connect Sync	Entra Cloud Sync
Users, groups, contacts	Yes	Yes
Device objects / hybrid join	Yes	No
Sign-in methods	PHS, PTA, federation	PHS only
Sync interval	30 min (floor)	~2 min
On-prem footprint	Server + SQL Express	Lightweight agent
Disconnected forests	No	Yes
Exchange hybrid, group writeback	Yes	No

For this build, Connect Sync is the only option. Cloud Sync does not synchronize device objects, and hybrid Microsoft Entra join requires the computer object to reach Entra ID. Choosing Cloud Sync would sync your users perfectly and make hybrid join structurally impossible.

Cloud Sync is the right choice for cloud-first environments where devices are Entra joined rather than hybrid joined, for merged or disconnected forests, and where a lighter on-premises footprint matters. Microsoft is investing in it as the long-term default, so expect the balance to shift.

Choose **Customize**, not Express. Express gives no control over sync scope, which is the setting that matters most.

Wizard page	Selection
Sign-in method	Password Hash Synchronization plus Enable single sign-on
Domain and OU filtering	Explicitly select OU=Employees , OU=Workstations and OU=Groups
Optional features	Leave defaults unless there is a specific requirement
Configure device options	Configure Microsoft Entra hybrid join , in this same sitting
Device operating systems	Windows 10 or later domain-joined
SCP configuration	Check the forest, Authentication Service = Microsoft Entra ID, supply Enterprise Admin credentials

WHY THE WORKSTATIONS OU MUST BE IN SCOPE

Hybrid Entra join requires the **computer object** to exist in Entra ID before a device can complete registration. If that OU is not synced, clients report `AzureAdJoined : NO` with `error_missing_device` — which looks like a client problem and is actually a sync-scope problem. Getting this right in the wizard saves a genuinely confusing troubleshooting session later.

9.4 STOP — verify the SCP before going any further

THIS IS THE SINGLE EASIEST STEP IN THE ENTIRE BUILD TO SKIP WITHOUT KNOWING

Configure device options is not part of the linear install wizard. It is a separate task chosen from the menu when Entra Connect is launched *again* after installation. The installer can complete, report success, and synchronize every user flawlessly — with hybrid join never configured. Nothing warns you. The failure only surfaces later, on a client, as `AzureAdJoined : NO`.

This has now cost time on two consecutive builds. Treat the check below as mandatory, not optional.

```
# On a domain controller
$scp = Get-ADObject -Filter 'Name -eq "62a0ff2e-97b9-4513-943f-0d221bd30080"' `
    -SearchBase "CN=Configuration,DC=ad,DC=contoso,DC=com" -Properties keywords
$scp.keywords
```

Required output — two values:

```
azureADName:yourtenant.onmicrosoft.com
azureADId:<your-tenant-guid>
```

Anything else — no object found, empty keywords, one value — means hybrid join was never configured. Relaunch Entra Connect, choose **Configure device options** → **Configure Microsoft Entra hybrid join**, and re-run this check before proceeding.

AFTER ANY CONFIGURATION CHANGE, RUN A FULL SYNC

`-PolicyType Delta` only processes objects that have *changed since the last cycle*. Your computer objects have not changed — what changed is the sync engine's rule set. A delta sync will appear to succeed and move nothing. Use `-PolicyType Initial` after every configuration change, including a repair.

9.5 Force and verify the first sync

```
Start-ADSyncSyncCycle -PolicyType Initial    # full sync
Get-ADSyncScheduler                        # confirm the 30-minute cycle is enabled
```

THE DEFAULT SYNC CYCLE IS 30 MINUTES

A large share of "it didn't work" during hybrid setup is actually "it hasn't synced yet." When testing, force the sync rather than waiting — and when something appears broken, check the clock before you check the configuration.

9.6 Licensing by group

Group-based licensing is three steps, and **all three are required**. Doing only the first two produces a licensing group that exists, has a SKU attached, and licenses nobody.

1. CREATE the group
Entra admin center > Groups > New group > Security > "LIC-M365-BusinessPremium"
2. ASSIGN the SKU to the group
M365 admin center > Billing > Licenses > select SKU > Assign licenses > select the group
3. PUT SOMEONE IN THE GROUP <-- the step that gets missed
Add the synced GG-AllEmployees group as a MEMBER of LIC-M365-BusinessPremium

STEP 3 IS THE ONE THAT BREAKS THE CHAIN

An empty licensing group throws no error and shows no warning. Users sync to Entra correctly, appear in the portal correctly, and are simply unlicensed. The symptom does not appear until much later and in a completely different place: **Intune enrollment silently fails**, because the MDM enrollment GPO runs in the user's context and an unlicensed user cannot enroll a device.

Diagnosing that backwards — from "the device isn't in Intune" to "the licensing group is empty" — is a long walk. Verify licensing at this phase instead.

Nest the on-premises role group into the cloud licensing group. This makes licensing automatic for every future user with no additional step:

```
GG-AllEmployees (synced from on-prem AD)
|
+--> member of --> LIC-M365-BusinessPremium (Entra, holds the SKU)
```

Your Phase 10 user-creation script already adds each new user to a **GG-** department group, which nests into **GG-AllEmployees**, which now nests into the licensing group. Create a user, and they are licensed on the next sync — no manual assignment, ever. This is the same AGDLP chaining logic you used for file permissions, applied to licensing.

WHY GROUP-BASED RATHER THAN PER-USER

Per-user assignment does not scale and drifts silently — you discover the gaps during an audit, or when someone cannot sign in. Group-based licensing makes onboarding "add to group" and offboarding "remove from group," and it is auditable at a glance. Requires Entra ID P1, which Business Premium includes.

Assigning a license *directly* to a user is still the correct **diagnostic** move when troubleshooting — it isolates whether the problem is the license or the group plumbing delivering it. That is not a contradiction of the group model. Just remove the direct assignment afterward, or you will have a user whose licensing survives removal from every group.

VERIFY BEFORE LEAVING THIS PHASE

```
M365 admin center > Billing > Licenses > select the SKU
Assigned count should equal your user count
Users should show as licensed VIA GROUP, not by direct assignment

M365 admin center > Users > Active users > [a user] > Licenses and apps
Confirm Intune is checked in the app list, not just the parent SKU
```

Also check **Licenses > Errors & Issues**. Group licensing failures — usually insufficient licenses or a conflicting service plan — surface there and nowhere else.

9.7 Entra dynamic device groups — create these now

```
Entra admin center > Groups > New group
Type: Security Membership type: Dynamic Device

DYN-Windows-Corporate:
(device.deviceOSType -eq "Windows") and (device.deviceOwnership -eq "Company")

DYN-Hybrid-Joined:
(device.deviceTrustType -eq "ServerAD")
```

WHY NOT JUST SYNC THE ON-PREMISES WORKSTATION GROUP

You cannot use a synced on-premises group to target devices in Intune. Computer objects arrive in Entra as *device* objects, and on-premises group membership of computers does not translate into Entra device group membership. Dynamic device groups are the answer: define the rule once and every matching device joins automatically, forever. See Part 2, Finding 10.

Phase 10 — Create the users

Script it from the first user, not the tenth. The script is the documentation.

```

Import-Module ActiveDirectory

$OU          = "OU=Employees,OU=Contoso,DC=ad,DC=contoso,DC=com"
$UPNSuffix   = "contoso.com"
$HomeShare   = "\\RC-DC01\Home"

function New-StrongPassword {
    $chars = 'abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ23456789!@#%'
    -join ((1..16) | ForEach-Object { $chars[(Get-Random -Maximum $chars.Length)] })
}

$Users = @(
    @{Sam="jsmith"; First="John"; Last="Smith"; Dept="Sales"; Group="GG-Sales"},
    @{Sam="sjohnson"; First="Sarah"; Last="Johnson"; Dept="Accounting"; Group="GG-Accounting"},
    @{Sam="mtorres"; First="Mike"; Last="Torres"; Dept="IT"; Group="GG-IT"}
)

$Report = foreach ($U in $Users) {
    if (Get-ADUser -Filter "SamAccountName -eq '$($U.Sam)'" -ErrorAction SilentlyContinue) {
        Write-Warning "$($U.Sam) already exists - skipping"; continue
    }
    $Password = New-StrongPassword
    try {
        New-ADUser -Name "$($U.First) $($U.Last)" `
            -GivenName $U.First -Surname $U.Last `
            -SamAccountName $U.Sam `
            -UserPrincipalName "$($U.Sam)@$UPNSuffix" `
            -DisplayName "$($U.First) $($U.Last)" `
            -Department $U.Dept `
            -Path $OU `
            -AccountPassword (ConvertTo-SecureString $Password -AsPlainText -Force) `
            -Enabled $true -ChangePasswordAtLogon $true `
            -HomeDrive "H:" -HomeDirectory "$HomeShare\$($U.Sam)"

        Add-ADGroupMember -Identity $U.Group -Members $U.Sam
        Write-Host "Created $($U.Sam)" -ForegroundColor Green
        [PSCustomObject]@{ User=$U.Sam; UPN="$($U.Sam)@$UPNSuffix"; Password=$Password }
    }
    catch { Write-Host "FAILED $($U.Sam): $_" -ForegroundColor Red }
}

$Report | Export-Csv "C:\Temp\NewUsers.csv" -NoTypeInformation
Write-Host "Credentials at C:\Temp\NewUsers.csv - deliver securely, then DELETE."

```

If PowerShell blocks the script with "not digitally signed":

```
Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass
```

Process scope reverts when the window closes. Do not set **Bypass** machine-wide on a client's server — **RemoteSigned** is the correct persistent setting if one is needed.

UNIQUE PASSWORDS, NOT A SHARED DEFAULT

Generating a unique password per account and forcing a change at first logon is barely more work than a shared default, and a shared default is a finding in every security review ever written. Build the habit where it costs nothing.

VERIFY

```

Get-ADUser -Filter * -SearchBase $OU -Properties UserPrincipalName, Department |
    Format-Table Name, SamAccountName, UserPrincipalName, Department

Start-ADSyncSyncCycle -PolicyType Delta      # then confirm they appear in Entra

```

Phase 11 — Build and join the workstations

By now the environment is complete. Onboarding should be mechanical.

1. Install Windows 11 **Pro or Enterprise**. Home cannot join a domain — verify the edition before promising anything.
2. **Unplug the network cable and disable Wi-Fi before starting OOBE**. Windows 11 forces a Microsoft account when it detects internet. With no network it offers a local account directly.

3. Complete OOBE with a **local** account. Reconnect the network afterward.
4. **Set the time zone.** Then verify the clock. See the callout below — this is not optional.
5. Patch fully. Install vendor drivers.
6. Rename to the planned hostname. Reboot.
7. Run the pre-join verification block below. Do not attempt the join until it passes.
8. Join the domain. The object lands in `OU=Workstations` automatically because of `redircmp`.
9. Reboot.
10. **Log in as a real domain user — never as an administrator.**

IF YOU ARE ALREADY ONLINE AT OOBE: SHIFT + F10

Press **Shift + F10** to open a command prompt during OOBE, then:

```
OOBE\BYPASSNRO
```

The machine reboots and OOBE now offers "I don't have internet." On builds where `BYPASSNRO` has been removed, use `start ms-cxh:localonly` from the same prompt instead.

11.1 Pre-join verification — run all four, in order

Every one of these is a hard prerequisite. Any failure means the join *will* fail, and the error Windows returns will not tell you which one.

```
# 1. DNS must point at a domain controller
ipconfig /all | Select-String "DNS Servers" -Context 0,2

# 2. SRV records must resolve – THIS is how a DC is located, not by ping
nslookup -type=SRV _ldap._tcp.dc._msdcs.ad.contoso.com

# 3. Clock skew must be under 5 minutes
w32tm /stripchart /computer:10.10.10.10 /samples:3 /dataonly

# 4. The DC must answer
Test-NetConnection 10.10.10.10 -Port 389
```

"AN ACTIVE DIRECTORY DOMAIN CONTROLLER COULD NOT BE CONTACTED" — DECODE IT

This message describes a symptom, not a cause, and it covers at least four unrelated failures. Work the list in order:

Real cause	How to confirm	Fix
Wrong DNS server	<code>ipconfig /all</code> shows the router or a public resolver	Fix DHCP option 6; never set client DNS by hand
SRV records unresolvable	<code>nslookup -type=SRV</code> returns nothing	DNS is wrong, or the zone lost its records — <code>net stop netlogon && net start netlogon</code> on the DC
Clock skew	<code>w32tm /stripchart</code> shows > 5 min offset	Set the time zone. Kerberos rejects the request; the client reports it as "no DC found"
Firewall / routing	<code>Test-NetConnection</code> on 389 or 445 fails	Permit 53, 88, 123, 135, 389, 445, 464 between the VLANs
Typo in the domain name	Re-read it character by character	Use the FQDN, not the NetBIOS name

Ping proves nothing here. A successful ping tests IP reachability using an address you supplied by hand. Domain join uses DNS SRV lookup and Kerberos, and both can fail completely on a network where ping works perfectly. Time zone in particular is nowhere in the error text and is a frequent cause on laptops, which ship configured for whatever region the image was built in.

STEP 8 DETERMINES DEVICE OWNERSHIP PERMANENTLY

The first user to sign in triggers MDM auto-enrollment under *their* identity. Signing in as `DOMAIN\Administrator` and manually adding a work account produces **Personal** ownership, reduced management capability, and application deployments that silently never install. This single shortcut caused days of unexplainable failure in the lab. Let the GPO do it.

STANDARD POST-JOIN VERIFICATION

```
whoami # DOMAIN\user
nltest /dsgetdc:ad.contoso.com # DC locator succeeds
gpresult /r # computer and user GPOs listed
net use # drive maps present
dsregcmd /status # DomainJoined : YES
```

11.2 Hybrid join — expect a delay, and know the sequence

AzureAdJoined : NO immediately after a domain join is **normal**. Hybrid join is a separate registration with a dependency chain, and each link takes time:

```
Domain join
-> computer object created in AD
-> Entra Connect syncs it to Entra ID (up to 30 min, or force it)
-> device object exists in Entra
-> client scheduled task registers the device
-> AzureAdJoined : YES
-> user logs off and back on
-> AzureAdPrt : YES (PRT issued at logon)
-> Intune enrollment can now proceed
```

Nothing downstream works until the PRT exists. Do not troubleshoot Intune before **AzureAdPrt : YES**.

IF AZUREADJOINED STAYS NO — WORK IT IN THIS ORDER

```
# 1. Read the actual error, at the bottom of the status output
dsregcmd /status # scroll to Diagnostic Data

# 2. On the DC: is the SCP present? (Phase 9.4)
$scp = Get-ADObject -Filter 'Name -eq "62a0ff2e-97b9-4513-943f-0d221bd30080"' `
-SearchBase "CN=Configuration,DC=ad,DC=contoso,DC=com" -Properties keywords
$scp.keywords

# 3. Force a FULL sync – delta will not move an unchanged object
Import-Module ADSync
Start-ADSyncSyncCycle -PolicyType Initial

# 4. Confirm the device object arrived
# Entra admin center > Devices > All devices (Pending is normal and fine)

# 5. Only then, on the client: clear stale state and re-register
dsregcmd /leave
Restart-Computer
# after reboot, logged in as the end user:
Get-ScheduledTask -TaskPath "\Microsoft\Windows\Workplace Join\" |
Start-ScheduledTask -TaskName "Automatic-Device-Join"
dsregcmd /status
```

READING THE DIAGNOSTIC DATA BLOCK

What it says	What it means
error_missing_device with Server operation: DeviceRenew	The client holds a cached device ID and is trying to renew a registration that no longer exists in Entra. Clear it with <code>dsregcmd /leave</code> and reboot — a plain retry will keep failing.
error_missing_device with Server operation: DeviceJoin	The device object genuinely is not in Entra. Sync scope or SCP problem.
0x80090303 on DEVICE_AUTO_KERB	Cloud Kerberos trust is not configured. Expected and harmless — the client falls back to <code>DEVICE_AUTO</code> . Not your problem.
AzureAdPrt : NO but AzureAdJoined : YES	Registration succeeded; the user has not logged on since. Sign out and back in.

Once hybrid join succeeds, confirm in Intune that the device appears within about 15 minutes and **Ownership shows Corporate**. If it shows Personal, fix it now: Devices, select the device, Properties, Device ownership, Corporate.

Phase 12 — Intune policy and applications

Enrollment is already handled by the Phase 8 GPO. Confirm the tenant side, then build policy in this order.

```
Entra admin center > Devices > Mobility (MDM and MAM) > Microsoft Intune
MDM user scope = All (or a pilot group)
```

Item	Location	Why in this order
Compliance policy	Devices, Compliance	Defines what "healthy" means. Everything downstream references it.
Conditional Access	Entra, Protection, Conditional Access	Require compliant device plus MFA. Exclude the break-glass account.
BitLocker	Devices, Configuration, Endpoint protection	Encryption with recovery key escrow to Entra
Update rings	Devices, Windows Update	Patch management with a pilot ring
Microsoft 365 Apps	Apps, All apps, Add, Microsoft 365 Apps	64-bit, Current Channel, Remove other versions = Yes. Assign to DYN - Windows - Corporate .
Defender for Endpoint	Endpoint security	Baseline antivirus and EDR policy

PORTAL PATHS CHANGE CONSTANTLY

Microsoft rearranges these admin centers regularly, and any written click-path — including this one — will go stale. Search the portal for the *feature name* ("Automatic enrollment", "Device settings", "Compliance policies") rather than following a path from a document. Distrusting a stale path is a correct instinct, not a failure to follow instructions.

KNOW WHICH DELIVERY MECHANISM YOU ARE TROUBLESHOOTING

Microsoft 365 Apps uses the Office CSP. Configuration profiles use OMA-DM. Packaged [.intunewin](#) Win32 apps, PowerShell scripts and remediations use the Intune Management Extension. The IME is *not* in the path for Microsoft 365 Apps, so its absence is never the cause of that app failing to install. Confirm the component is actually in the path before spending time on it.

12.1 Verifying an app deployment from the client side

The Intune portal reports installation status on its own schedule and lags reality by a long way. Check the client directly — you get an answer in seconds instead of waiting on a console that may not update for an hour.

```
# Is the Office installer actually running?
Get-Process OfficeClickToRun, setup, OfficeC2RClient -ErrorAction SilentlyContinue |
    Select-Object Name, StartTime, CPU

# Is the Intune Management Extension present? (Win32 apps and scripts only)
Get-Service IntuneManagementExtension -ErrorAction SilentlyContinue

# What did the MDM channel actually do?
Get-WinEvent -LogName "Microsoft-Windows-DeviceManagement-Enterprise-Diagnostics-Provider/Admin" `
    -MaxEvents 30 | Select-Object TimeCreated, Id, LevelDisplayName, Message | Format-List

# Did Office land?
Get-ItemProperty "HKLM:\SOFTWARE\Microsoft\Office\ClickToRun\Configuration" -ErrorAction SilentlyContinue |
    Select-Object ProductReleaseIds, VersionToReport, UpdateChannel
```

[OfficeClickToRun.exe](#) consuming CPU means the deployment fired and is downloading. That is a successful outcome — you do not need to wait for the portal to confirm it.

WHEN AN ASSIGNMENT APPEARS TO DO NOTHING

1. **Licensing** — the user must have an Intune license. This is the most common cause by a wide margin.
2. **Group membership** — a dynamic device group can take several minutes to populate after a device registers. Open the group and confirm the device is actually in it.
3. **Targeting type** — a device-targeted assignment installs per-machine; a user-targeted one installs at user logon. Mixing them up produces "nothing happened."
4. **Sync** — force one from the client: Settings, Accounts, Access work or school, select the account, Info, Sync.
5. **PRT** — `dsregcmd /status` must show `AzureAdPrt : YES`. Without it the device cannot authenticate to Intune at all.

Group Policy and Intune conflicts

When both systems target the same setting the result is not reliably predictable. Decide per category which system owns what, and write the decision down.

Group Policy owns	Intune owns
Drive mappings and Folder Redirection	Security baselines
On-premises resource access	BitLocker and encryption
Legacy line-of-business requirements	Windows Update rings
Logon banners, local group membership	Application deployment

Phase 13 — Second domain controller and real redundancy

Adding DC02 is the easy part. Making the environment actually fail over is the part that gets skipped.

DOMAIN CONTROLLERS DO NOT LIVE IN YOUR SERVERS OU

You built `OU=Servers` in Phase 5, and no domain controller will ever appear in it. That is correct behavior, not a mistake.

Promoting a server to a domain controller **moves its computer object into the built-in Domain Controllers OU**, and it stays there. That OU is where the Default Domain Controllers Policy is linked, and that policy grants the user rights a DC needs to function. Moving a DC out of it breaks authentication in ways that are difficult to diagnose.

`OU=Servers` is for **member servers** — file servers, print servers, application servers, an Entra Connect host, a management server. In a single-server lab it will legitimately sit empty. Build it anyway: it costs nothing and the structure is correct the moment a second server exists.

13.1 Build DC02

- Independent hardware, or a VM on an **always-on** hypervisor host. Not a desktop that gets powered off.
- If virtualized on Hyper-V, attach to an **External** virtual switch. The Default Switch is NAT and cannot reach the LAN.
- Disable host time synchronization before promotion:**

```
# On the Hyper-V host
Get-VMIntegrationService -VMName "DC02" -Name "Time Synchronization" | Disable-VMIntegrationService
```

- Base build as Phase 1. Static IP. Preferred DNS = DC01, Alternate = `127.0.0.1`.
- Join to the domain as a member server first**, reboot, then promote. Direct promotion of a non-joined server fails more often, and the credential error it produces is misleading.
- Add the AD DS role, Promote, **Add a domain controller to an existing domain**, DNS server checked, Global Catalog checked, DSRM password to the vault, replicate from DC01.
- Install management tools: `Install-WindowsFeature GPMC, RSAT-AD-Tools, RSAT-DNS-Server`
- Set time to follow the domain hierarchy:

```
w32tm /config /syncfromflags:domhier /update
Restart-Service w32time
w32tm /query /source # must NOT say "VM IC Time Synchronization Provider"
```

13.2 Wire up the redundancy — the part that is usually missed

Component	Required configuration
DC01 DNS client settings	Preferred: DC02 , Alternate: <code>127.0.0.1</code>
DC02 DNS client settings	Preferred: DC01 , Alternate: <code>127.0.0.1</code>
DHCP option 6	Both DC addresses, in order
DHCP service itself	DHCP failover between the two DCs, or a split scope
File shares	Still single-homed on DC01. Use DFS Namespaces and DFS Replication if file service redundancy is required.
FSMO roles	Documented. Seizure procedure written. See SOP-07.

```
# DNS client settings, cross-pointing
Set-DnsClientServerAddress -InterfaceAlias "Ethernet" -ServerAddresses ("10.10.10.11","127.0.0.1") # on DC01
Set-DnsClientServerAddress -InterfaceAlias "Ethernet" -ServerAddresses ("10.10.10.10","127.0.0.1") # on DC02

# DHCP failover
Add-DhcpServerv4Failover -Name "DC01-DC02" -PartnerServer "RC-DC02.ad.contoso.com" -ScopeId 10.10.10.0 -LoadBalancePercent 50
```

A SECOND DC DOES NOT CREATE REDUNDANCY BY ITSELF

If clients have only one DNS server address they cannot locate the second DC when the first is down, because they cannot resolve the `_ldap._tcp` SRV records that find a domain controller. Logins may still appear to succeed on **cached credentials**, which makes the naive failover test produce a false pass. Fix DNS delivery first, then test with an account that has never logged into that machine. See Part 2, Findings 13 and 14.

VERIFY REPLICATION, THEN VERIFY FAILOVER PROPERLY

```
Get-ADDomainController -Filter * | Format-Table Name, HostName, Site, IsGlobalCatalog
repladmin /replsummary # no errors, no large deltas
repladmin /showrepl
dcdiag /v
dcdiag /test:dns
```

Then run the full failover procedure from Part 2, Finding 14, including the never-before-logged-in account.

Phase 13.5 — Azure Arc (optional)

Azure Arc projects on-premises servers into Azure as managed resources, giving you a single inventory, Azure Policy compliance, Update Manager, and Log Analytics across on-prem and cloud. It is genuinely useful in MSP work and it is the natural pairing with the hybrid identity you just built.

PREREQUISITES THAT ARE NOT OBVIOUS

- **An Azure subscription.** A Microsoft 365 subscription is not one. They are separate billing constructs.
- **Azure RBAC, which is separate from Entra roles.** Being Global Administrator in Entra grants you *nothing* in Azure. You must explicitly elevate access or have an existing Owner assign you a role at the subscription scope. This surprises nearly everyone the first time.
- The **Azure Connected Machine Onboarding** or **Contributor** role at the target resource group.

13.5.1 Grant Azure access to your admin account

```
# Signed in as the account that created the tenant / subscription owner
Azure portal > Subscriptions > [subscription] > Access control (IAM)
  > Add role assignment > Contributor > assign to your daily admin account

# If NO account has access at all:
Entra admin center > Properties
  > Access management for Azure resources = Yes (elevates the Global Admin temporarily)
  > then assign a proper RBAC role and set this back to No
```

ENTRA ROLES AND AZURE RBAC ARE TWO DIFFERENT SYSTEMS

Entra ID roles (Global Administrator, Intune Administrator) govern the *directory* — users, groups, devices, tenant settings. Azure RBAC roles (Owner, Contributor, Reader) govern *Azure resources* — subscriptions, resource groups, VMs. They are deliberately separate so that directory administration and cloud spend can be delegated independently.

The practical consequence: a brand-new Global Administrator often cannot see a single Azure subscription, and it looks like a bug. It is not. Grant RBAC explicitly at the subscription or resource group scope, and prefer **Contributor** over **Owner** for a working account — Owner can also grant access to others, which is rarely needed day to day.

13.5.2 Onboard the servers

1. Azure portal → **Azure Arc** → **Machines** → **Add** → Generate script
2. Choose resource group, region, OS = Windows
3. Run the generated script on each server **as administrator**

azcmagent is not recognized. The installer adds the agent to the system PATH, but your current PowerShell session started before that happened. Open a new elevated session, or call it by full path:

And a PowerShell trap worth internalizing: the backtick is a line-continuation character and must be the **last character on the line**. A single trailing space after it breaks the command with a confusing parse error. When a pasted multi-line command fails for no visible reason, collapse it to one line before troubleshooting anything else.

The machine appears as Connected in Azure Arc → Machines.

None of this is interesting. All of it is what keeps you on the account.

- System State backup of every DC, nightly, written off the server
- File data backed up with versioning and an offsite copy
- 3-2-1: three copies, two media types, one offsite
- **Test a restore** and record the date you tested it

- Network topology diagram, one page, current
- IP and hostname register
- AD structure: OUs, groups, and *what each group grants*
- GPO register: name, link, purpose, last reviewed
- Credential vault entries, complete, including DSRM and break-glass
- Runbooks — the SOPs in Part 4
- Licensing register: what is owned, what it entitles, renewal dates
- Evaluation and certificate expiry dates, with calendar reminders

- LAPS deployed and verified retrievable
- Separate accounts: daily user, admin account, break-glass. Never browse the internet from a DC.
- MFA on every administrator
- Domain Admins membership reviewed quarterly — it should be nearly empty
- Remote access via RMM or VPN. **Never** RDP exposed to the internet.
- A Disabled OU with a defined retention period

- Backup success or failure alerting — the single most valuable alert you can configure
- Disk space on every server volume
- AD replication health
- Certificate and license expiry
- Failed logon spikes and account lockout patterns

Part 4 — Standard Operating Procedures

These are the repeatable tasks. Written so someone else could run them, which is the actual test of whether a procedure is documented or just remembered.

SOP-01 — New user onboarding

Trigger: HR or the client requests a new account. Do not start without a written request naming the person, start date, department, and manager.

1. Confirm a license is available before creating anything. Running out mid-onboarding is avoidable embarrassment.
2. Create the account in `OU=Employees` using the Phase 10 script or its single-user equivalent. Correct UPN suffix, department field populated, change password at next logon set.
3. Add to the correct `GG-` role group. Nothing else — role group membership drives everything downstream.
4. Add to the licensing group (`LIC-M365-*`).
5. Force a sync: `Start-ADSyncSyncCycle -PolicyType Delta`. Wait, then confirm the user appears in Entra with the right UPN.
6. Confirm the license landed in the Microsoft 365 admin center.
7. Deliver credentials by a channel separate from the username. Never both in the same email.
8. Assign or prepare a workstation per SOP-03.
9. Record in the ticket: account name, UPN, groups, license, workstation asset tag.

Verify: the user logs in on their assigned machine, drives map, Documents redirect, and they can sign in to Microsoft 365 on the web.

SOP-02 — User offboarding

Trigger: written notice from an authorized contact. This is the SOP clients audit. Get it in writing, every time, and timestamp each step.

1. **Disable the account.** Do not delete it — deletion destroys mailbox and file ownership chains.
2. **Reset the password** to something random. Disabling alone does not invalidate an existing session.
3. **Revoke active sessions:** `Revoke-MgUserSignInSession -UserId user@contoso.com`. Without this a signed-in session can persist for hours.
4. Remove from all `GG-` role groups. Record what they were in the ticket before removing.
5. Remove from the licensing group only after mailbox handling is decided.
6. Convert the mailbox to shared, or delegate access to the manager, per the client's policy.
7. Move the account to `OU=Disabled`.
8. Set out-of-office or mail forwarding if requested — in writing.
9. Retrieve the device. Wipe or reassign per SOP-03.
10. Remove the device from Intune and Entra if it is being retired.
11. Retrieve or transfer files from the home folder and redirected folders.
12. Note the retention date. Delete the account only after the client's retention period expires.

Verify: sign-in blocked, sessions revoked, group membership empty, device accounted for.

SOP-03 — Workstation deployment

1. Record the asset tag, serial number, model and assigned user in the asset register before touching it.
2. Add the serial number to Intune corporate device identifiers if using that method.
3. Install or reset Windows 11 Pro or Enterprise. Complete OOBE with a **local** account.
4. Patch fully. Install vendor drivers and firmware.
5. Rename to the naming convention. Reboot.
6. Confirm DHCP delivered both DCs as DNS.
7. Join the domain. Reboot.
8. **Log in as the assigned end user.** Not as an administrator, and not as yourself.
9. Wait for MDM auto-enrollment. Confirm in Intune within 15 minutes.
10. **Confirm Ownership shows Corporate.** Fix it immediately if not.
11. Confirm BitLocker is on and the recovery key is escrowed to Entra.
12. Run the verification block below. Hand over only when it is clean.

```
gpresult /r
net use
dsregcmd /status
manage-bde -status C:
```

SOP-04 — New file share or permission change

1. Get the request in writing, naming who needs access and at what level.
2. Create or identify the Domain Local resource group: `DL-<Resource>-<Access>`.
3. Nest the appropriate Global role group into it. **Never add a user directly.**
4. If a new share: create the folder, share it with `Authenticated Users` Full Control, enable Access-Based Enumeration.
5. Set NTFS using the standard model: Domain Admins and SYSTEM Full Control, the `DL-` group at Modify or Read, CREATOR OWNER Full Control on subfolders and files only.
6. Add the drive mapping to the Drive Maps GPO with item-level targeting on the `GG-` group, Action = Update.
7. Test with a real account from the group. Test that a non-member **cannot see the folder at all.**
8. Update the share and permission register.

BEFORE TOUCHING AN EXISTING SHARE

Export the current ACL first: `Get-Acl D:\Path | Export-Clixml C:\Temp\acl-backup.xml`. Never tick "Replace all child object permission entries" on a production tree without knowing exactly what is beneath it.

SOP-05 — Add a domain controller

1. Confirm the existing directory is healthy first: `dcdiag /v` and `repadmin /replsummary` must be clean. Never add a DC to a sick domain.
2. Build the server per Phase 1. If virtual, disable host time synchronization.
3. Static IP. Preferred DNS = an existing DC, Alternate = `127.0.0.1`.
4. **Join as a member server. Reboot.** Then promote.
5. Add AD DS role, promote as an additional DC, DNS and Global Catalog checked, DSRM password to the vault.
6. Install GPMC and RSAT tools on it.
7. Set `w32tm /config /syncfromflags:domhier /update`.
8. Cross-point DNS client settings on all DCs.
9. Add the new DC to DHCP option 6 on every scope.
10. Configure DHCP failover if it will serve DHCP.
11. Update the topology diagram and IP register.

```
Get-ADDomainController -Filter * | Format-Table Name, HostName, IsGlobalCatalog
repadmin /replsummary
dcdiag /test:dns
```

SOP-06 — Monthly maintenance check

Thirty minutes per client per month. This is what turns reactive break-fix into managed services.

```
# Directory health
dcdiag /v
repadmin /replsummary
Get-ADDomainController -Filter * | Format-Table Name, IsGlobalCatalog
netdom query fsmo

# Time
w32tm /query /status
w32tm /query /source

# Backup
Get-WBSummary

# Storage
Get-Volume | Where-Object DriveLetter | Format-Table DriveLetter, SizeRemaining, Size

# Security review
Get-ADGroupMember "Domain Admins" | Select-Object Name
Search-ADAccount -AccountDisabled -UsersOnly | Measure-Object
Search-ADAccount -AccountInactive -TimeSpan 90.00:00:00 -UsersOnly | Select-Object Name

# Licensing and expiry
slmgr /dlv
```

Also check: Intune non-compliant devices, Entra Connect sync errors, failed backup alerts, patch compliance, and any certificate expiring within 60 days.

SOP-07 — Domain controller failure and FSMO seizure

If the DC is recoverable: repair it and let replication converge. Do not seize roles. Seizing is one-way.

If the DC is permanently gone:

1. Confirm it is truly unrecoverable. Seizure cannot be undone and the old DC must never be brought back online afterward.
2. Identify which roles it held — from your documentation, or `netdom query fsmo`.
3. Seize the roles onto a surviving DC:

```
Move-ADDirectoryServerOperationMasterRole -Identity "RC-DC02" -OperationMasterRole
PDCEmulator,RIDMaster,InfrastructureMaster,SchemaMaster,DomainNamingMaster -Force
```

The `-Force` switch performs a seizure when the current holder is unreachable.

4. Remove the dead DC's metadata: delete its object in Active Directory Users and Computers, its NTDS Settings in AD Sites and Services, and its records in DNS.
5. If it was the PDC emulator, reconfigure the external time source on the new holder (Phase 2.3).
6. Update DHCP option 6 and all DNS client settings to remove the dead DC's address.
7. Verify: `netdom query fsmo`, `dcdiag /v`, `repadmin /replsummary`.
8. Build a replacement DC per SOP-05.

SOP-08 — Account lockout triage

The most common ticket in any AD environment, and usually not a forgotten password.

1. Confirm the lockout and find the source:

```
Search-ADAccount -LockedOut | Select-Object Name, LastLogonDate

# On the PDC emulator, find where the bad attempts came from
Get-WinEvent -FilterHashtable @{LogName='Security'; ID=4740} -MaxEvents 20 |
  Format-List TimeCreated, Message
```

2. Event 4740 names the calling computer. That is your actual culprit.
3. Common real causes, in order of frequency: a phone or tablet with a stale saved password; a mapped drive using cached credentials; a scheduled task or service running as the user; a stale RDP session on another machine; Credential Manager entries.
4. Unlock and reset only after finding the source. Otherwise it locks again in minutes and you have taught the user that IT does not fix things.
5. Clear stale credentials on the offending device: `cmdkey /list` then `cmdkey /delete:<target>`.

Part 5 — Assessing and Troubleshooting an Existing Network

Most of your work will not be greenfield. It will be inheriting an environment built by someone who left, with no documentation, where something is broken and the client wants it fixed today.

The discipline is: **find out what is actually there before you change anything**. The temptation on day one is to start fixing. Resist it. You cannot safely change a system you have not mapped, and the thing that looks broken is often load-bearing.

5.1 Discovery — run this before touching anything

Run from a domain-joined machine with administrative credentials. Capture the output to a file and keep it. It is the "before" picture and it will be the most valuable thing you produce in week one.

```
# ----- FOREST AND DOMAIN -----
Get-ADForest | Format-List Name, ForestMode, DomainNamingMaster, SchemaMaster, GlobalCatalogs
Get-ADDomain | Format-List Name, DNSRoot, NetBIOSName, DomainMode, PDCEmulator, RIDMaster, InfrastructureMaster
netdom query fsmo

# ----- DOMAIN CONTROLLERS AND HEALTH -----
Get-ADDomainController -Filter * | Format-Table Name, HostName, IPv4Address, Site, IsGlobalCatalog, OperatingSystem
dcdiag /v
dcdiag /test:dns
repadmin /replsummary
repadmin /showrepl

# ----- TIME -----
w32tm /query /status
w32tm /query /source
w32tm /monitor

# ----- DNS -----
Get-DnsServerZone
Get-DnsServerForwarder
Get-DnsServerScavenging

# ----- DHCP -----
Get-DhcpServerInDC
Get-DhcpServerv4Scope
Get-DhcpServerv4OptionValue -ScopeId <scope> # check option 6 (DNS)

# ----- STRUCTURE -----
Get-ADOrganizationalUnit -Filter * | Select-Object Name, DistinguishedName
Get-ADDomain | Select-Object ComputersContainer, UsersContainer # was redircmp ever run?

# ----- GROUP POLICY -----
Get-GPO -All | Sort-Object DisplayName | Format-Table DisplayName, GpoStatus, ModificationTime
Get-GPOReport -All -ReportType Html -Path C:\Temp\AllGPOs.html

# ----- PRIVILEGED ACCESS -----
Get-ADGroupMember "Domain Admins" | Select-Object Name, objectClass
Get-ADGroupMember "Enterprise Admins" | Select-Object Name
Get-ADGroupMember "Schema Admins" | Select-Object Name

# ----- ACCOUNT HYGIENE -----
Search-ADAccount -AccountInactive -TimeSpan 90.00:00:00 -UsersOnly | Select-Object Name, LastLogonDate
Search-ADAccount -PasswordNeverExpires -UsersOnly | Select-Object Name
Search-ADAccount -AccountInactive -TimeSpan 90.00:00:00 -ComputersOnly | Select-Object Name

# ----- FILE SERVICES -----
Get-SmbShare | Where-Object { $_.Name -notlike "*" } | Format-Table Name, Path, FolderEnumerationMode

# ----- HYBRID IDENTITY -----
Get-ADSyncScheduler
Get-ADSyncConnector | Select-Object Name, Type
```

5.2 What you will actually find, and what to do about it

These are the findings that turn up repeatedly in SMB environments. Prioritized by risk, not by how easy they are to fix.

Pri	Finding	What it means and what to do
1	No tested backup of AD or file data	Nothing else matters if this is missing. Fix first, then test a restore and document the date.
1	RDP exposed to the internet	Primary ransomware entry vector. Close it today. Move to VPN or RMM. Escalate immediately if you find it.
1	Single domain controller	Authentication, DNS and often DHCP all fail together. Highest-value upsell and easiest to justify.
1	Shared local admin password everywhere	One machine compromised equals all machines compromised. Deploy LAPS.
2	Domain Admins with ten or more members	Usually every IT person's daily account plus service accounts. Separate daily from privileged. Aim for near-empty.
2	Users with permissions applied directly to folders	Unauditable and undocumentable. Migrate to group-based over time; do not rip it out in one pass.
2	Everyone group with Full Control on shares	Move to the standard model. Audit what is actually being accessed before restricting.
2	No MFA on administrators	Enable it. Create a break-glass account first, excluded from Conditional Access.
2	PDC emulator syncing from Local CMOS Clock	No external time anchor. Fix per Phase 2.3.
3	Computer objects sitting in <code>CN=Computers</code>	Those machines receive no policy. Run <code>redircmp</code> and move the existing objects.
3	Everything configured in the Default Domain Policy	Untroubleshootable. Split into purpose-named GPOs gradually; document before changing.
3	Dozens of GPOs, unlinked or unnamed	Generate the full HTML report, map what is actually linked, then archive the orphans.
3	<code>.local</code> domain name	Note it, do not rename. Add a routable UPN suffix when cloud services are introduced.
3	Entra Connect installed on a domain controller	Note as a finding. Plan migration to a member server.
3	AD Recycle Bin not enabled	Enable it. Free, immediate, no downside.
4	Disabled accounts never cleaned up	Establish a Disabled OU and a retention policy.
4	No reverse DNS zone or scavenging	Create the zone, enable scavenging. Stale records cause intermittent, confusing failures.
4	Server evaluation licenses in production	Check <code>slmgr /dlv</code> . Plan the conversion before expiry — it requires demotion on a DC.

DISCIPLINE ON AN INHERITED NETWORK

Document first, change second, and change one thing at a time. If you change five things and something breaks, you have five suspects and no way to bisect. Also: assume anything odd is load-bearing until proven otherwise. The strange scheduled task or the share nobody can explain is frequently how payroll runs.

5.3 Triage decision trees

"I can't log in"

1. Is it one user or many? Many means infrastructure. One means account.
2. One user: `Search-ADAccount -LockedOut`. If locked, go to SOP-08 and find the source before unlocking.
3. Account fine? Try a different machine. Works elsewhere means it is the machine, not the account.
4. Many users, one machine: check the machine's secure channel — `Test-ComputerSecureChannel -Repair`.
5. Many users, many machines: check DNS, then the DCs, then time.

```
nltest /dsgetdc:ad.contoso.com
w32tm /query /status          # skew over 5 minutes breaks Kerberos everywhere
dcdiag /v
```

6. Check whether a DC is down and whether clients have a second DNS server to fall back to.

"I can't get to the file share"

1. Can they reach it by UNC path? `\\SERVER\Share`. If yes, the problem is the drive mapping, not access.
2. Can they see it but not open files? That is NTFS, not share permissions.
3. Can they open but not save? Modify permission missing, or the "Applies to" scope is wrong.
4. Can nobody reach it? Check the server, the SMB service, and the firewall.

5. Check effective access: folder Properties, Security, Advanced, Effective Access tab. Test with the actual user account.
6. Confirm group membership actually resolved on the client: `whoami /groups` . A recently added group requires a full logoff to take effect.

"Group Policy isn't applying"

1. `gpresult /h C:\gpo.html` and read it. This answers most of these tickets on its own.
2. Is the GPO listed as applied, or denied? Denied tells you it is a filtering problem.
3. Not listed at all: is the GPO linked to an OU that actually contains this object? Check where the object lives, not where you think it lives.
4. Is the object in `CN=Computers` or `CN=Users` ? No GPO reaches there.
5. Is the link enabled? Is the GPO status set to enabled for the half you need?
6. Security filtering: does the object have Read and Apply?
7. User settings on a computer OU, or vice versa? Check which half of the GPO you configured.
8. Folder Redirection or drive maps: did the user fully log off, or just lock the screen?

"The device isn't showing up in Intune"

1. `dsregcmd /status` . `DomainJoined : YES` and `AzureAdJoined : YES` are both required.
2. `AzureAdJoined : NO` : the computer object is not in Entra. Check Entra Connect sync scope includes the workstation OU, then force a full sync.
3. Both YES but not in Intune: check the user has an Intune license and that MDM user scope includes them.
4. Check the MDM auto-enrollment GPO is applying to the machine.
5. In Intune, check **Ownership**. Personal means it was enrolled by hand and will behave badly. Change it to Corporate.
6. App not installing: confirm the assignment targets a group that actually contains this device. On-premises groups cannot contain Entra devices — use a dynamic device group.

Part 6 — Appendices

Appendix A — Command reference

On a domain controller

Command	What it tells you
<code>dcdiag /v</code>	Overall DC health. Run after any change and during every maintenance check.
<code>dcdiag /test:dns</code>	DNS-specific health, including SRV record registration
<code>repadmin /replsummary</code>	Replication status at a glance. Any error here is urgent.
<code>repadmin /showrepl</code>	Detailed per-partner replication
<code>netdom query fsmo</code>	Which DC holds each of the five roles
<code>w32tm /query /status</code>	Current time source, stratum, and skew
<code>w32tm /monitor</code>	Time offset across every DC in the domain
<code>Get-ADDomainController -Filter *</code>	Every DC, site, and Global Catalog status
<code>Get-SmbShare</code>	Shares, paths, and whether ABE is on
<code>Get-ADSyncScheduler</code>	Entra Connect sync cycle state
<code>Start-ADSyncSyncCycle -PolicyType Delta</code>	Force an incremental sync
<code>Start-ADSyncSyncCycle -PolicyType Initial</code>	Force a full sync
<code>slmgr /dlv</code>	Licensing status and remaining evaluation days

On a client

Command	What it tells you
<code>gpresult /h C:\gpo.html</code>	The readable policy report. Use this one, not <code>/r</code> , for real troubleshooting.
<code>gpresult /r</code>	Quick list of applied GPOs
<code>whoami /groups</code>	Whether group membership actually resolved in this session
<code>dsregcmd /status</code>	Domain join, Entra join, and MDM enrollment state in one place
<code>nltest /dsgetdc:<domain></code>	Which DC the client found, and how
<code>net use</code>	Current mapped drives
<code>klist / klist purge</code>	Kerberos tickets; purge to force a fresh token
<code>Test-ComputerSecureChannel -Repair</code>	Repairs a broken machine account trust relationship
<code>manage-bde -status C:</code>	BitLocker state
<code>ipconfig /all</code>	What DNS servers DHCP actually handed out
<code>cmdkey /list</code>	Cached credentials — a frequent cause of repeat lockouts

Appendix B — Error to cause to fix

Symptom	Actual cause	Fix
Domain join fails, domain not found	Client DNS not pointing at a DC	Fix DHCP option 6. Never set client DNS statically.
GPO absent from <code>gpresult</code>	Linked to an OU that does not contain the object	Link where the user or computer object actually lives
New PC receives no policy	Object landed in <code>CN=Computers</code>	<code>redircmp</code> , then move existing objects
Can browse share, cannot open files	NTFS permissions	Grant Modify to the resource group; verify the "Applies to" scope
Total access loss after a cleanup	<code>SYSTEM</code> or admin removed from the ACL	Take ownership, rebuild from the standard model

<code>gpupdate /force</code> hangs	Drive map action = Replace with the drive in use	Change action to Update
Folder Redirection Event 502	Home and redirected roots collide; folder not owned by the user	Separate the roots
Drives appear for the wrong people	Item-level targeting on the wrong group	Fix targeting; enable ABE so folders hide too
<code>AzureAdJoined : NO , error_missing_device</code>	Computer OU not in Entra Connect sync scope	Add OU to scope, full sync, <code>dsregcmd /leave</code> then <code>/join</code>
Intune apps never install	Device ownership = Personal from manual enrollment	Change ownership to Corporate; enroll via GPO going forward
Intune assignment does nothing	Targeted an on-prem group that cannot contain Entra devices	Use an Entra dynamic device group
Second DC promotion, bad credentials	Name resolution, IPv6 registration, or time hierarchy	Join as a member server first, fix the time source, then promote
Virtualized DC with odd Kerberos errors	Taking time from the hypervisor host	Disable the Time Synchronization integration service; <code>syncfromflags:domhier</code>
Arc agent shows not connected	Expired onboarding token	<code>azcmagent disconnect --force</code> then <code>connect</code>
<code>azcmagent</code> not recognized	PATH not refreshed in the current session	New elevated session, or call by full path
Multi-line PowerShell fails oddly	Whitespace after a backtick	Collapse to a single line
Script blocked, not digitally signed	Execution policy	<code>Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass</code>
GPMC fails when a DC is down	Console targeting the offline DC	Change Domain Controller; then fix DNS redundancy properly
VM cannot reach the LAN	Attached to the Hyper-V Default Switch, which is NAT	Attach to an External virtual switch
Host loses internet after creating a vSwitch	Host now routes through the new vEthernet adapter with no DNS set	Set DNS on the vEthernet adapter
Account locks out repeatedly	Cached credentials on a phone, mapped drive, or service	Event 4740 names the source device; clear with <code>cmdkey</code>
"A domain controller could not be contacted" — but ping works	Clock skew over 5 min (often a wrong time zone), or DNS not pointing at a DC	<code>w32tm /stripchart</code> and <code>nslookup -type=SRV</code> . Ping tests neither.
"The name has already been shared" (error 2118)	<code>New-SmbShare</code> only creates; the share exists	<code>Set-SmbShare</code> to modify, or <code>Remove-SmbShare</code> then re-create
Oobe will not offer a local account	Windows 11 detects internet	Disconnect the network, or Shift+F10 then <code>OOBE\BYPASSNRO</code>
Full sync run but nothing moved	Used <code>-PolicyType Delta</code> after a config change	<code>-PolicyType Initial</code> — delta only processes changed objects
<code>Start-ADSyncSyncCycle</code> not recognized	ADSync module not loaded; session predates the install	<code>Import-Module ADSync</code> in a new session
Device in Entra but never enrolls in Intune	User has no Intune license — often an empty licensing group	Confirm the licensing group has members; check Licenses > Errors & Issues
Global Admin cannot see any Azure subscription	Entra roles do not grant Azure RBAC	Assign Contributor at subscription scope, or elevate access temporarily
Random client loses network / duplicate IP	DHCP pool overlaps the static infrastructure range	Add an exclusion range, or start the pool above the static block
Domain controller missing from <code>OU=Servers</code>	Not a fault — DCs live in the built-in Domain Controllers OU	Leave it. Moving a DC out breaks its default policy.

Appendix C — Port reference

Port	Service	Needed for
53 TCP/UDP	DNS	Everything. Check this first, always.
88 TCP/UDP	Kerberos	Authentication
123 UDP	NTP	Time sync — and therefore Kerberos
135 TCP	RPC endpoint mapper	AD replication, remote management
389 TCP/UDP	LDAP	Directory queries
445 TCP	SMB	File shares, Group Policy delivery from SYSVOL
464 TCP/UDP	Kerberos password change	Password resets
636 TCP	LDAPS	Secure directory queries
3268 / 3269 TCP	Global Catalog	Forest-wide lookups
3389 TCP	RDP	Remote Desktop. Never open to the internet.
5985 / 5986 TCP	WinRM	PowerShell remoting
49152–65535 TCP	Dynamic RPC	AD replication. The range firewalls usually block by accident.

Appendix D — Pre-flight checklists

Before promoting a domain controller

- Server fully patched and rebooted
- Renamed to its final hostname
- Static IP configured, DNS pointing correctly
- Correct time zone, clock accurate within one minute
- If virtual: host time synchronization disabled
- If additional DC: already joined as a member server, existing directory verified healthy
- DSRM password chosen and recorded in the vault

Before joining the first client

- OU structure built, `redircmp` and `redirusr` run
- Security groups created, both tiers
- Shares created and permissioned, ABE enabled, tested with a test account
- All GPOs built and validated against one test user and one test machine
- UPN suffix added before any real user was created
- DHCP handing out the correct DNS servers

Before declaring a build complete

- System State backup running and a restore tested
- AD Recycle Bin enabled
- External time source configured on the PDC emulator
- LAPS deployed and a password successfully retrieved
- Failover tested correctly — never-before-used account, DC01 off
- Documentation package complete: diagram, registers, credentials, runbooks
- Monitoring and backup alerting configured
- Licensing and evaluation expiry dates on a calendar

Appendix E — Open items carried forward

State at the end of the Round 2 build. Ordered by value, not effort.

#	Task	Notes
1	Complete Phase 13.2 redundancy wiring	DC02 is promoted but the environment still assumes one DC. Cross-point DNS, add DC02 to DHCP option 6, configure DHCP failover.
2	Run the failover test correctly	Part 2, Finding 14. Use an account that has never logged into the test machine, or cached credentials will produce a false pass.
3	Fix the DHCP pool overlap	Pool currently spans the whole subnet including the gateway and both DCs. Start it above the static block.
4	Correct the DHCP search domain	Appliance field still set to <code>lan</code> while option 15 says the AD domain. Make them agree.
5	Clean up stale AAAA records	Router advertisements are off but public IPv6 records persist in the zone. Phase 2.2.
6	Test the System State restore	Backup is scheduled and running. It is a hypothesis until a restore has been tested.
7	Nest GG-AllEmployees into the licensing group	Replaces the direct per-user license assignments made during troubleshooting. Remove the direct assignments afterward.
8	Remove the license from the break-glass account	Break-glass should be sterile — no mailbox, no daily use, no licensed services beyond what sign-in requires.
9	Intune compliance policy and Conditional Access	Devices are enrolled but nothing defines or enforces a healthy state. Exclude break-glass from all CA.
10	Verify LAPS end to end	GPO exists. Retrieve an actual password to prove the pipeline works.
11	Build a management workstation with RSAT	Stop administering from the DC console.
12	Start the Appendix G toolkit	Begin with <code>New-LabUsers.ps1</code> . One script per session.
13	Produce the documentation package	Diagram, IP register, GPO register, credential vault. The portfolio deliverable.
14	Check evaluation expiry on both DCs	<code>slmgr /dlv</code> . Cannot convert while a DC — plan ahead.

Superseded — the original Round 1 list, for reference

#	Task	Notes
1	System State backup on both DCs	Currently no recovery path exists at all
2	External NTP on the PDC emulator	Phase 2.3. Forest has no time anchor.
3	Disable Hyper-V time sync on DC02, set <code>syncfromflags:domhier</code>	Currently a competing stratum-1 time source
4	Move client DNS to DHCP, listing both DCs	Without this the second DC provides nothing
5	Cross-point DC DNS client settings	DC01 to DC02, DC02 to DC01, each with itself as alternate
6	Re-run the failover test correctly	Part 2, Finding 14. Use an account that has never logged in.
7	Enable AD Recycle Bin	One command, permanent benefit
8	Deploy Windows LAPS	Phase 8.2C
9	Fix WS-SALES01 device ownership	Change Personal to Corporate in Intune, or unenroll and re-enroll via GPO
10	Create Entra dynamic device groups	Replaces the manually maintained Entra group
11	Build a management workstation with RSAT	Stop administering from the DC console
12	Check evaluation expiry on both DCs	<code>slmgr /dlv</code> . Cannot convert while a DC — plan ahead.
13	Move DC02 to always-on hardware, or demote when idle	A DC that is off half the time is a liability
14	Intune compliance policy and Conditional Access	Create the break-glass account first
15	Produce the documentation package	Diagram, registers, runbooks. The portfolio deliverable.

Appendix F — Rebuild-from-scratch drill

The most useful thing to do with this playbook is to prove you do not need it.

Wipe the lab. Rebuild it from Part 3 alone — no AI assistance, no searching, timed. Target for a second pass: Phases 0 through 11 in one working day.

If you can do it, you own this material rather than having followed it. If you stall, the exact point where you stall identifies the concept you were pattern-matching rather than understanding. Both outcomes are useful and the second is more useful.

Things to change on the rebuild, so it is a new problem rather than a repetition:

- Use a routable domain name, not `.local`
- Put Entra Connect on a member server rather than the DC
- Build both domain controllers before joining any client
- Serve DHCP from Windows rather than the firewall
- Deploy LAPS and backup as part of the build, not afterward
- Write the documentation as you go rather than at the end

A NOTE ON WHAT "FINISHED" MEANS

Environments are never finished. A lab you are still breaking and extending is a better interview story than one that was declared complete and switched off. The value of this build was never the final state — it was the eighteen findings in Part 2, every one of which came from something not working.

Appendix G — Reusable Script Templates

Typing a forty-line PowerShell block into a console for every client is not sustainable, and it is a syntax error waiting to happen. One mistyped character in a long block means retyping it. This appendix is the fix.

The principle: write the logic once, keep the client-specific values outside the logic, and never edit the script itself again.

G.1 Directory layout

Keep this on a USB stick, in OneDrive, or in a private Git repository. Every client gets a config file; the scripts are identical everywhere.

```
Toolkit\
├── config\
│   ├── contoso.psd1
│   └── homebiz.psd1
├── scripts\
│   ├── New-LabOUStructure.ps1
│   ├── New-LabGroups.ps1
│   ├── New-LabFileShares.ps1
│   └── New-LabUsers.ps1
└── input\
    └── users-homebiz.csv
```

G.2 The client config file

A `.psd1` is a PowerShell data file — a hashtable, no executable code. It is the only file you edit per client.

```
# config\homebiz.psd1
@{
    DomainDN      = "DC=ad,DC=homebiz,DC=beer"
    OrgOU         = "Homebiz"
    UPNSuffix     = "homebiz.beer"
    NetBIOS       = "HOMEBIZ"
    FileServer    = "HL-DC01"
    ShareRoot     = "S:"
    HomeShare     = "\\HL-DC01\Home"
    Departments   = @("Sales", "Accounting")
}
```

Load it with one line:

```
$Cfg = Import-PowerShellDataFile ".\config\homebiz.psd1"
$Cfg.UPNSuffix      # homebiz.beer
```

G.3 A parameterised script

```
# scripts\New-LabUsers.ps1
[CmdletBinding(SupportsShouldProcess)]
param(
    [Parameter(Mandatory)][string]$ConfigPath,
    [Parameter(Mandatory)][string]$UserCsv,
    [string]$OutputPath = "C:\Temp\NewUsers.csv"
)

Import-Module ActiveDirectory
$Cfg = Import-PowerShellDataFile $ConfigPath
$OU = "OU=Employees,OU=$( $Cfg.OrgOU ),$( $Cfg.DomainDN )"

function New-StrongPassword {
    $chars = 'abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ23456789!@#%'
    -join ((1..16) | ForEach-Object { $chars[(Get-Random -Maximum $chars.Length)] })
}

$Report = foreach ($U in (Import-Csv $UserCsv)) {

    if (Get-ADUser -Filter "SamAccountName -eq '$($U.Sam)'" -ErrorAction SilentlyContinue) {
        Write-Warning "$($U.Sam) already exists - skipping"
        continue
    }

    $Password = New-StrongPassword

    if ($PSCmdlet.ShouldProcess($U.Sam, "Create AD user")) {
        try {
            $Params = @{
                Name           = "$($U.First) $($U.Last)"
                GivenName      = $U.First
                Surname        = $U.Last
                SamAccountName = $U.Sam
                UserPrincipalName = "$($U.Sam)@($Cfg.UPNSuffix)"
                DisplayName     = "$($U.First) $($U.Last)"
                Department     = $U.Dept
                Path            = $OU
                AccountPassword = (ConvertTo-SecureString $Password -AsPlainText -Force)
                Enabled        = $true
                ChangePasswordAtLogon = $true
                HomeDrive      = "H:"
                HomeDirectory   = "$($Cfg.HomeShare)\$($U.Sam)"
            }

            New-ADUser @Params
            Add-ADGroupMember -Identity $U.Group -Members $U.Sam
            Write-Host "Created $($U.Sam)" -ForegroundColor Green

            [PSCustomObject]@{
                User      = $U.Sam
                UPN       = "$($U.Sam)@($Cfg.UPNSuffix)"
                Password  = $Password
            }
        }
        catch { Write-Host "FAILED $($U.Sam): $_" -ForegroundColor Red }
    }
}

$Report | Export-Csv $OutputPath -NoTypeInformation
Write-Host "`nCredentials at $OutputPath - deliver securely, then DELETE." -ForegroundColor Yellow
```

The input file:

```
# input\users-homebiz.csv
Sam,First,Last,Dept,Group
jsmith,John,Smith,Sales,GG-Sales
sjohnson,Sarah,Johnson,Accounting,GG-Accounting
```

Running it:

```
# Dry run first - shows what would happen, changes nothing
.\scripts\New-LabUsers.ps1 -ConfigPath .\config\homebiz.psd1 `
    -UserCsv .\input\users-homebiz.csv -WhatIf

# For real
.\scripts\New-LabUsers.ps1 -ConfigPath .\config\homebiz.psd1 `
    -UserCsv .\input\users-homebiz.csv
```

G.4 Why each piece is there

Element	What it buys you
<code>param()</code> with <code>[Parameter(Mandatory)]</code>	PowerShell prompts for anything missing rather than failing halfway through
<code>SupportsShouldProcess</code>	Gives you <code>-WhatIf</code> free. Dry-run any destructive script before committing. Use this on every script that writes.
Splatting (<code>@Params</code>)	Kills the backtick continuation problem entirely. No line-continuation characters, no trailing-whitespace failures, and the parameters stay readable.
CSV input	Onboarding 40 users is the same effort as onboarding 2. The client can even fill in the CSV.
Existence check + <code>continue</code>	Idempotent — safe to re-run after a partial failure
<code>try/catch</code> per item	One bad row does not abort the other 39
Config file separate from logic	New client = one new <code>.psd1</code> . The scripts never change, so they never accumulate client-specific bugs.

G.5 Getting them to run

```
# Unblock files downloaded or copied from another machine
Get-ChildItem .\scripts\*.ps1 | Unblock-File

# Allow local scripts for this session only - reverts when the window closes
Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass
```

Do not set `Bypass` machine-wide on a client's server. `RemoteSigned` is the correct persistent setting if one is needed.

NEXT STEP WHEN YOU ARE READY

Put the toolkit in a private Git repository. You get version history, the ability to roll back a change that broke something, and a portfolio artifact a hiring manager can actually look at. "Here is my deployment toolkit" is a much stronger interview moment than "I know PowerShell."

G.6 Build order for your own toolkit

Do not try to write all of these at once. Convert one playbook phase per session, starting with the ones you type most often.

Order	Script	Covers
1	<code>New-LabUsers.ps1</code>	Phase 10 — longest to type, most repeated
2	<code>New-LabGroups.ps1</code>	Phase 6 — both group tiers plus nesting
3	<code>New-LabOUStructure.ps1</code>	Phase 5 — OUs plus <code>redircmp</code> / <code>redirusr</code>
4	<code>New-LabFileShares.ps1</code>	Phase 7 — folders, shares, ABE, NTFS
5	<code>Test-LabHealth.ps1</code>	The Part 5 discovery block, wrapped as one command

That last one is the sleeper. A single script that runs every health check and dumps a timestamped report is the thing you will actually use every week once you are working an MSP queue.